New Deck
Work in Progress 2024
For OAE

DÆDÆLUS

Chiplet Mesh Networking

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Plus the DÆ Gang: Steve, Dugan, Sahas, Sam, Roger & Susan

Base Presentation Deck- Technology Stack for Applied Graph Theory

Making Transactions Reliable

Distributed Applications suffer from Silent Data *Loss* and Data *Corruption*

Everyone has their theory; but no one really knew why. *Until Now!*

Quantum or Classical?

Quantum physics deals with systems whose states are entangled and hidden until measured

DÆDÆLUS' protocols causally intertwine the state machines on either side of a link via a ***bidirectional interaction*** of within the FPGA's which is hidden from the x86 Host Processor until queried

The mathematics and combinatorics of quantum entanglement and superposition (i.e. Spekkens Model) provide a robust framework for designing transaction algorithms which are deterministic, atomic, and ***reversible***

Applications Need an API to Negotiate with the Database

Big Deal

Database Read-Committed Snapshot Isolation (RCSI) and other Experimental Isolation Semantics

Lingua Franca

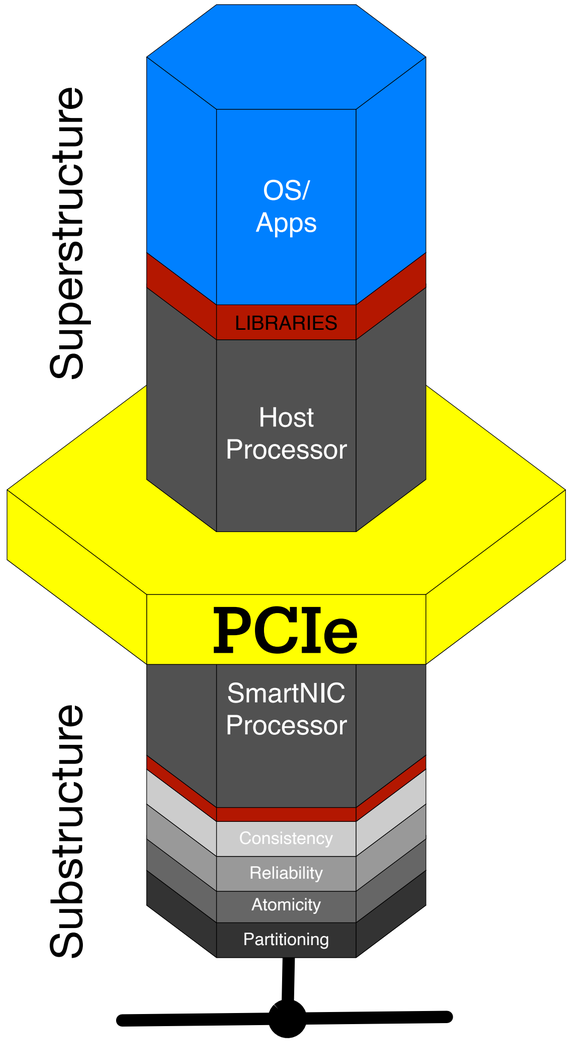
Reactors Mechanisms to Verify or Expose Non-Confluent Behavior

Æthernet

Shannon Information (stirred and not shaken)

Antithesis Network Systems Simulator

DÆDÆLUS



DÆ

DÆDÆLUS

Unreliable Transactions Cause Silent Data Loss & Corruption

We solve fundamental problems the industry believes are impossible: liveness, transactions and consensus — *without timeouts and retries* — which lead to Cascade Slowdowns: Retry Storms, Metastable Failures, Limpware, Transaction Failure, and Silent Data Corruption

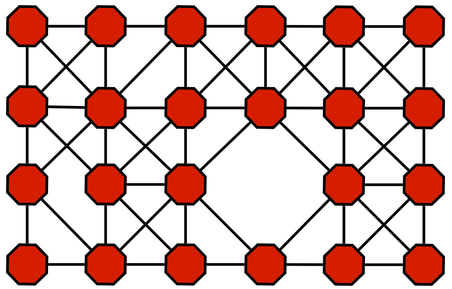
Chiplet Mesh Networking Can Solve This Problem at Layer 2
But it needs a compatible API at Layer 7

Benefit: CAP, CALM, and Exactly Once Semantics For ACID Transactions

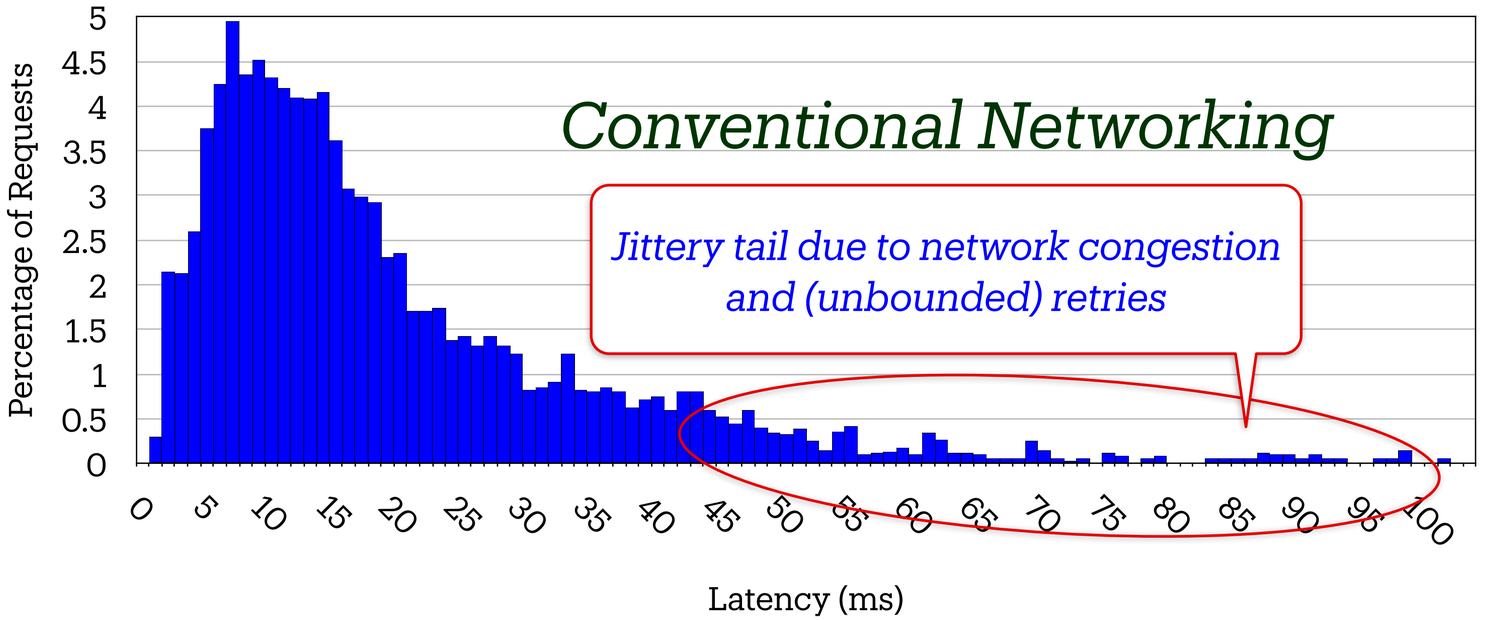
Tail Latency Before & After

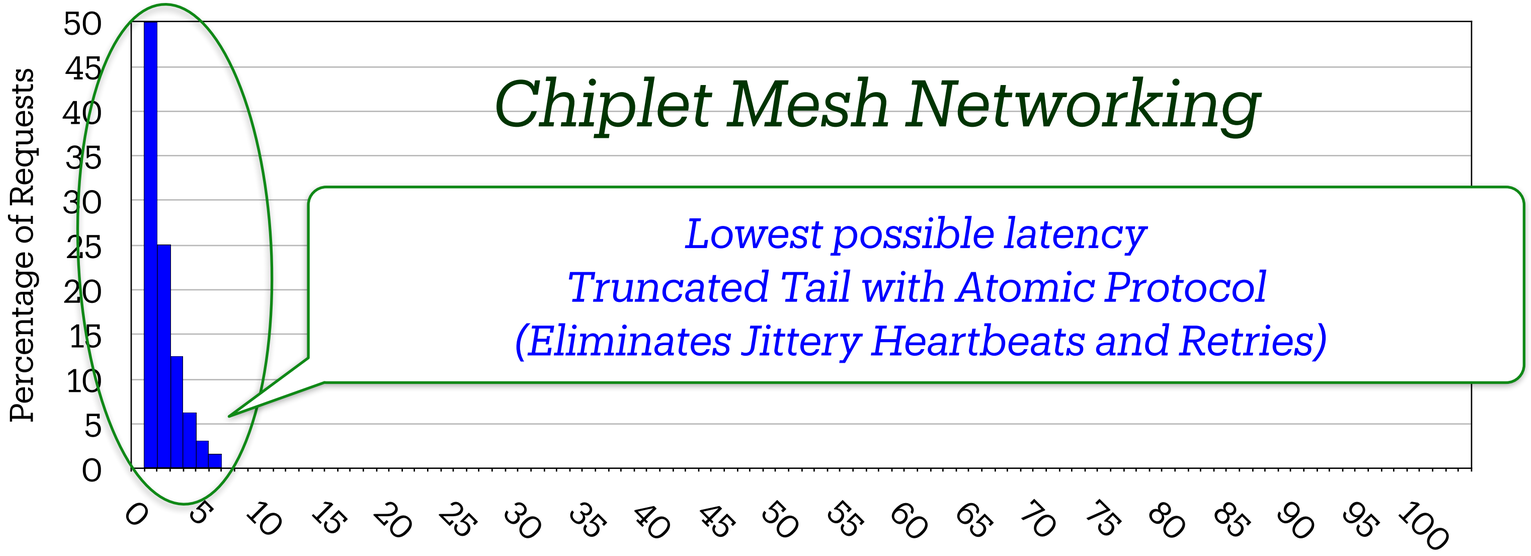
DÆDÆLUS reduces latency in ways conventional networks cannot:

- Direct connections
- Atomic Multicast, in parallel over 8 ports, instead of serial over 1
- Truncated Tail Latency — protocol knows if it failed or succeeded (without heartbeats or timeouts)



Chiplet Mesh





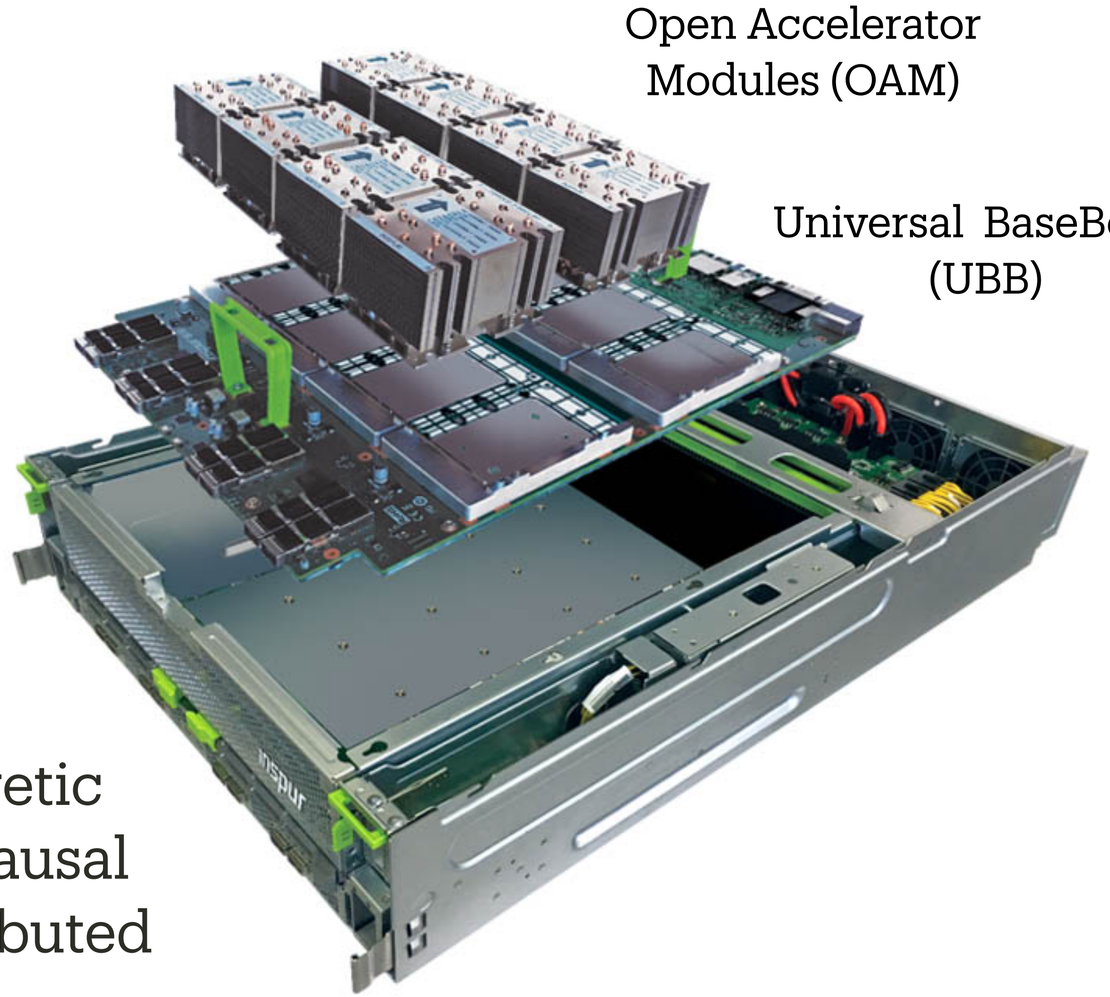
Chiplet Motherboards are Miniature Cities

Chiplet Motherboards are Miniature Cities

Physical Stuff

Bare Metal Reality

- Baseboard
- 8 Modules
  - Palm Size Computers, Cube Heatsinks
- Matches DÆ 8 port Octagonal Mesh
- Bare Metal: What exists after manufacture, and before power is applied?
- Abstractions include: physical network, logical network and virtual network
- Topmost: Digital Twin, an information theoretic model which represents the temporal and causal behavior of the system, against which distributed systems programmers can program

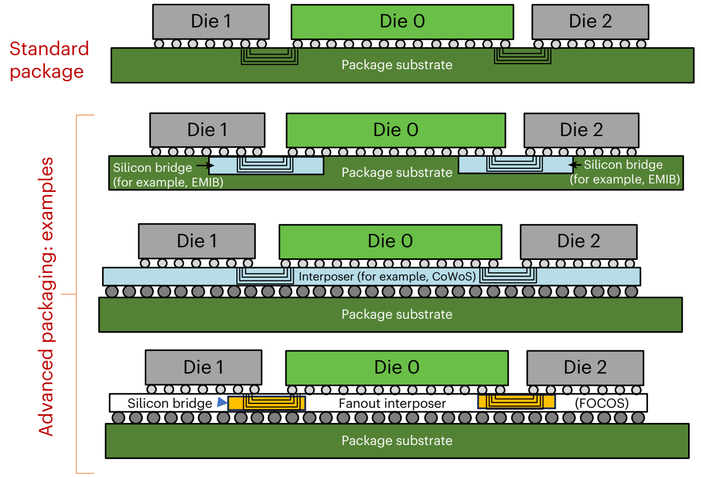


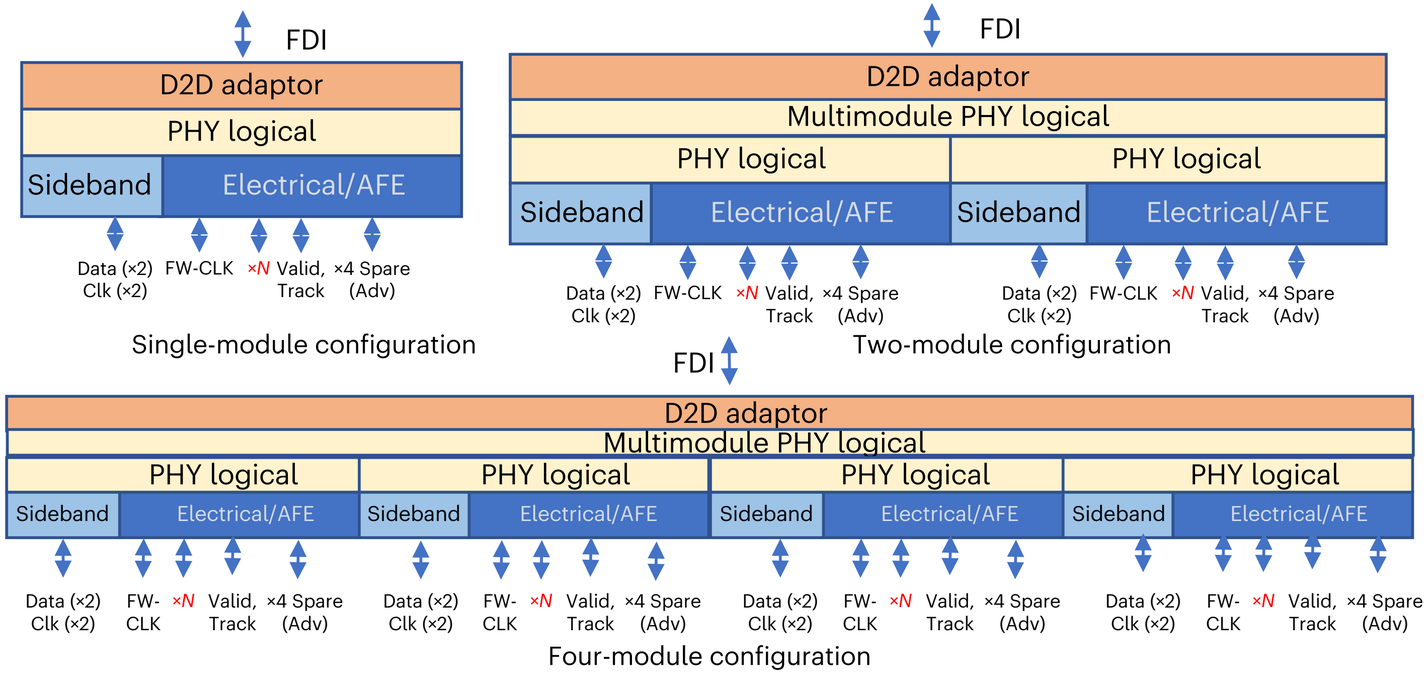
Going Deeper

Chiplet Infrastructure

- From UCIe Nature-Electronics Paper
- How do we map extensible (stacked) graphs?
- Virtual on top of Logical
- Logical on top of Physical
- Physical at the bottom

- To create a “Digital Twin”
- Start with the Pictures
- Map to Algebra
- Program in Pictures





Digital Twin Perspective

- We usually associate the three usual dimensions with length, width and height (in any order), and the fourth dimension with time. However, mathematically, dimensions are essentially an artificial construct and do not have a fixed meaning linked to any specific aspect of reality. By associating each dimension with a variable, they can be used to represent pretty much anything that we can put a parameter to. ... Representations of any dimension are thus used to model various aspects of geographic information, such as the typical geographic coordinates, but also time and scale. <Ken Arroyo Ohori>

- Spatial information describes the location of objects in space and the relationships between them using abstracted digital representations of real-world entities such as terrain, cities, roads and buildings
- We can use similar techniques to model Datacenters, racks, baseboards, modules, chips and interconnects. BUT ...
- The biggest issue in infrastructure management is not in managing spatial relationships, it's managing *temporal* relationships. Timestamps are fraught with theoretical and practical issues that have been swept under the rug by Computer Scientists, resulting in ***silent data loss***

DÆDÆLUS Digital Twins

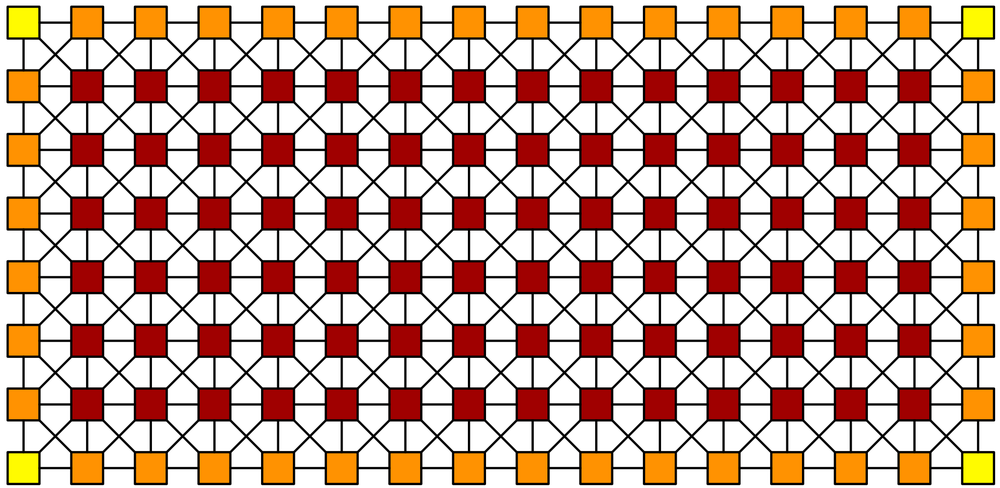
- Instead of relying on timestamps and synchronized clocks (which cannot be synchronized anyway because of Special Relativity, and Indefinite Causal Order in Quantum Mechanics)
- We create a ‘conserved quantities’ ***resource*** in each Ethernet Link, to guarantee temporal properties that are otherwise impossible to recover higher up in the protocol stack
- Timestamps remain useful as a *label*, for HFT, IOT & Sensor Networks

- **Entanglement:** classically emulate the No-Cloning Theorem; with *precise* injection and extraction of **Shannon information** at Layer 2
- Revolutionizes distributed systems programming. Guarantees **Exactly Once Semantics,** for ACID Transactions
- Resource is implemented in Network Accelerators as IP Blocks, invoked by a parameterized **Conserved Quantities** Transaction API in host processors (TPI)

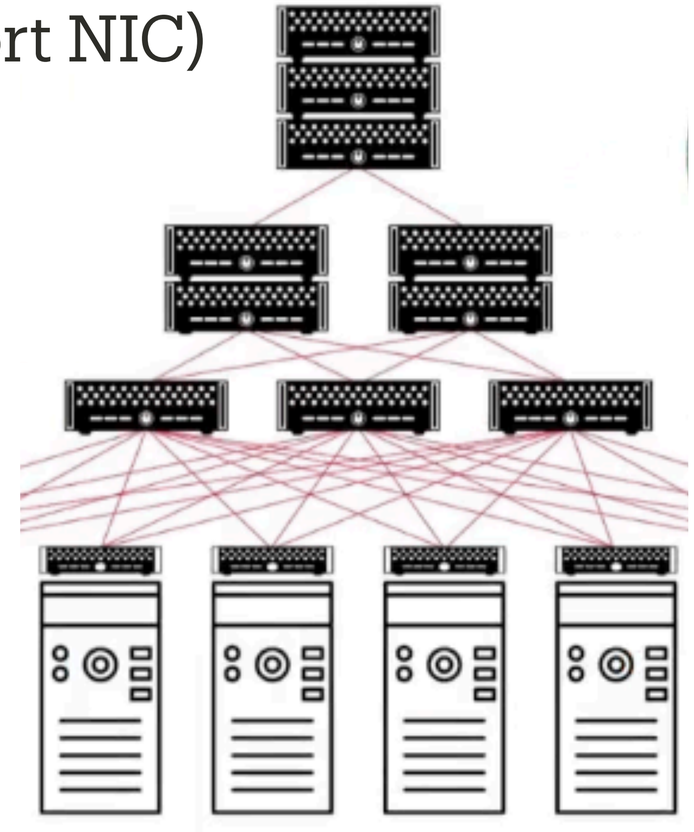
Topology: Chiplet vs. Racks

Manage distributed systems software with Graphs, not Lists

- Comparison: Latency, Cost, Bandwidth, Power - Chiplets do far better
- Indistinguishable Nodes (Host Processor + 8-port NIC)
- Use all 8 lanes of the NIC independently
- Clos ‘combines’ multiple lanes for Bandwidth

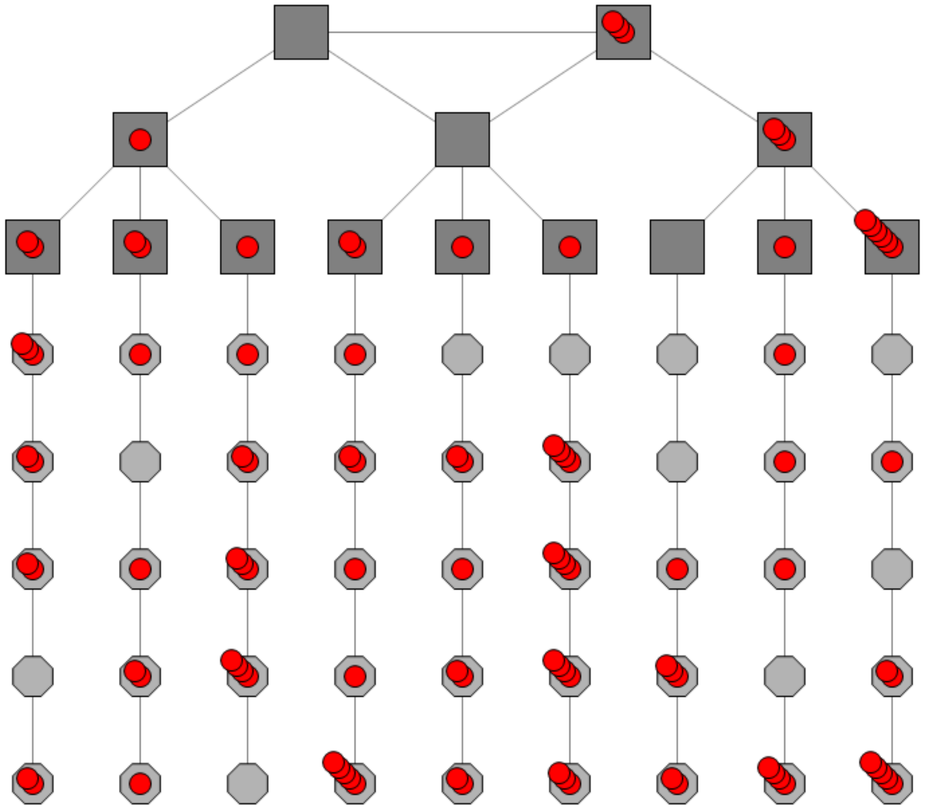


Chiplet Architecture



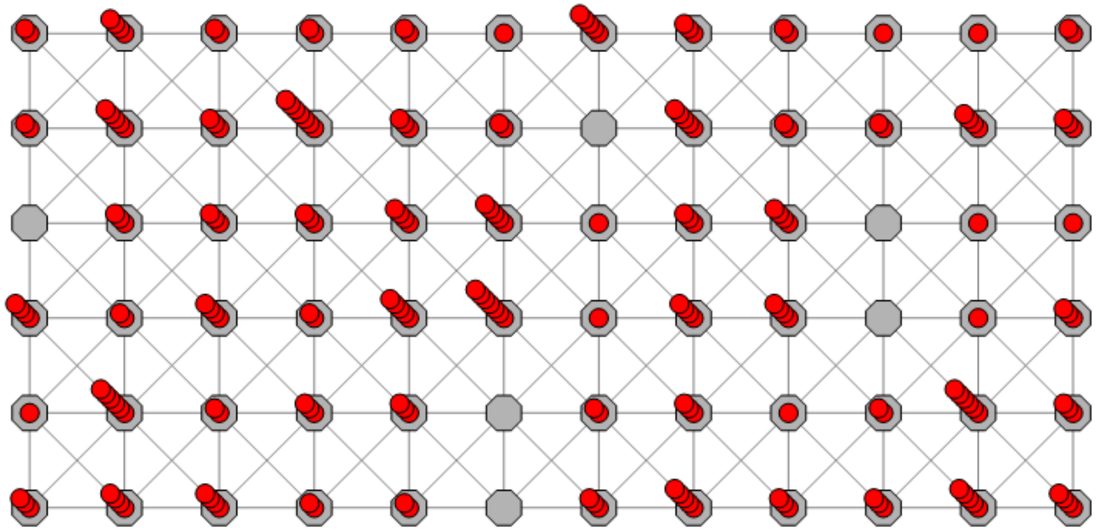
Rack Architecture

Logical Stuff: Digital Twin



Rack (Clos) Topology

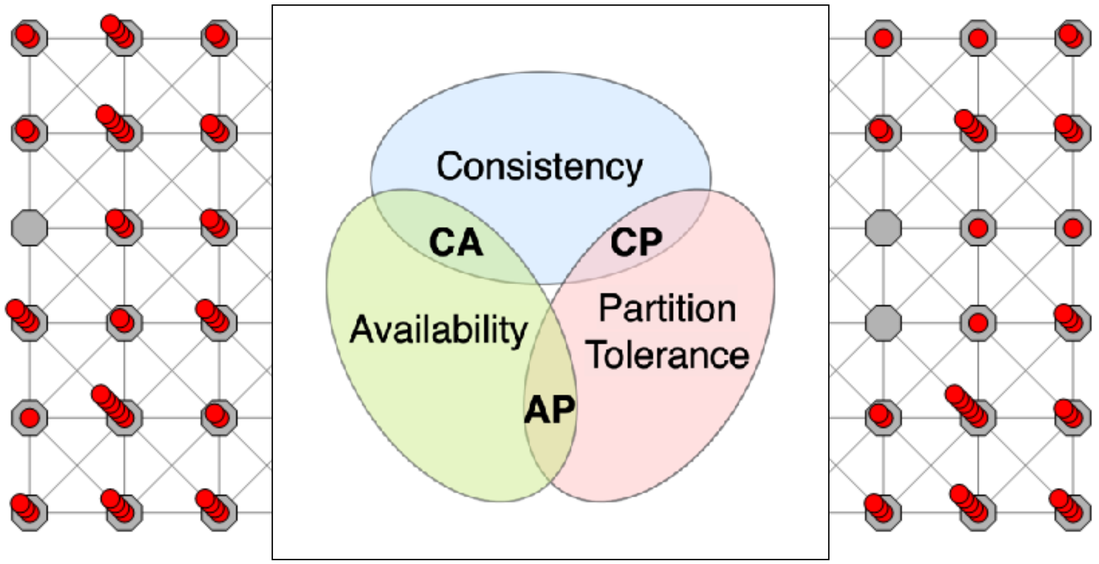
Compare Bandwidth, Latency,
Packet Loss, and ‘Temporal Intimacy’



Chiplet Universal Motherboard

For Distributed Applications

- **Eliminating CAP Theorem Tradeoffs**
  — means eliminating partitions
- **Chiplet Mesh;**
  - **Detect**, isolate and heal failures, on 8-port routing trees, with local information only, *in microseconds*
  - **Enable** tenant reconfigurations, in *milliseconds*
  - **Observe** ‘on Demand’ for distributed systems debugging



We make networks *appear* unbreakable

Product Hypothesis

- **Best practice servers** deployed by the million today are a local optimum: *a more cost-effective design center exists:*
  - Server modules on a 2D, 2.5D, 3D, or 3.5D *tiled plane*
  - Each tile connects only to its immediate neighbors
  - To Software, this is a ***topology*** management problem
- **Can a Software Defined 'Chiplet Infrastructure' make it easier to:**
  - Bring-up — repair and *retrain* interconnects *under software control?*
  - Monitor — Observe, Provision, Account for … Tenant usage?
  - Deploy — Highly Dynamic Graph-based Computations?
  - Protect — From *all* Known Perturbations?
  - Transact — Guarantee: *Determinism on Demand?*

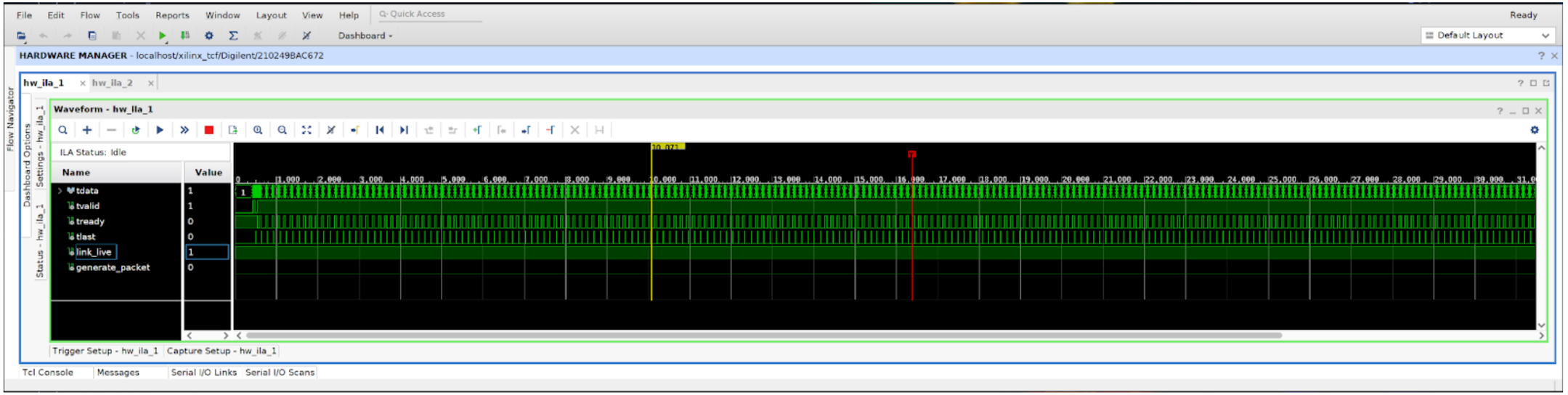
Chiplet Mesh Networking

Imagine an Ethernet (or UCIe) Protocol that:

- Has minimum possible buffering (and hence jitter)
- Always pairs (never drops acknowledgements)
- Guarantees transaction sequences (order)
- Maintains Conserved Quantities in the Link
- Heals in less than a microsecond; without skipping a beat

- How would that affect your distributed or ML applications?
- What new thing would be enabled?
- What impossible thing would now become possible?

It Works: Bits on the Wire Demo

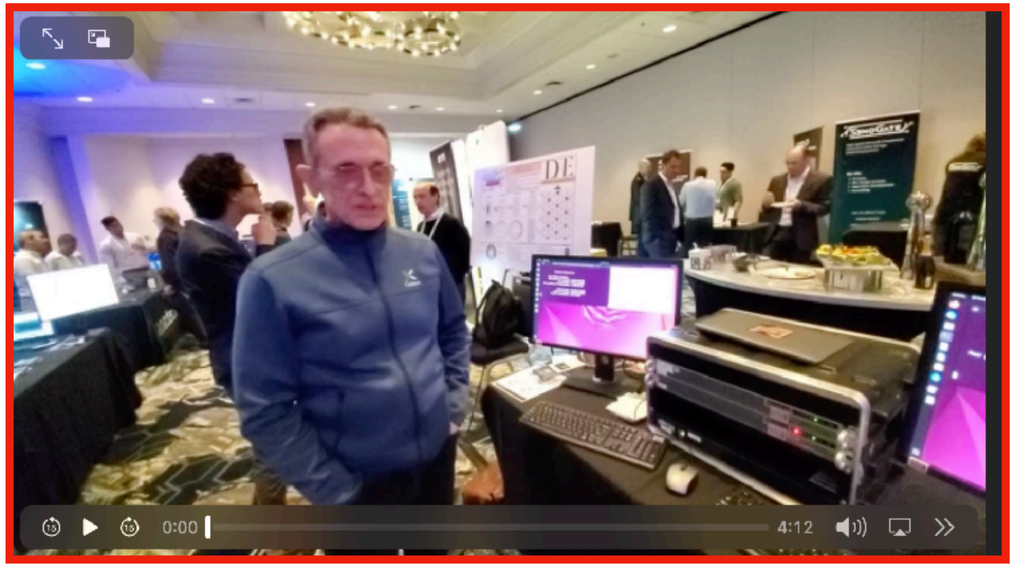


Integrated Logic Analyzer trace of two link partners transferring information. Round trip time 1.2uS

Chiplet Summit
DÆDÆLUS Booth #219

Chiplet Mesh Networking Industry Association

for Distributed System Developers



Product: Hyperdata IP Blocks (Licensed)

Subtransactions in L1, Accessible from Transaction API

- We can't recover lost information (events) at L3 (IP) or L4 (TCP) — *when we dropped them at L2*

- ***Hyperdata*** must be carried in the L1 packet (it can't be thrown away at L2, or L3; this is how data structures get silently corrupted)
- ***Hyperdata*** defines *bounded* forward & reverse sequences for subtransaction(s)
- Metadata & data are the payload, which may safely maintain copies on both ends, *without* maintaining 'complementarity bits' for exactly once semantics

Product: Discovery IP Blocks (Licensed)

Applied Graph Theory independently enables Physical, Logical and Virtual layers to manage computational resources for Microdatacenters

- Everything becomes APPLIED GRAPH THEORY

- **The Scouting Protocol: Theseus**
  - Discovers neighborhoods for Chiplet based Microdatacenters
- **The Routing Protocol: Ariadne**
  - Connects neighborhoods using Hypergraph provisioning and facilitates rapid reconfiguration after perturbations

* Larry Peterson's Message pairs & John Ousterhout's Homa.

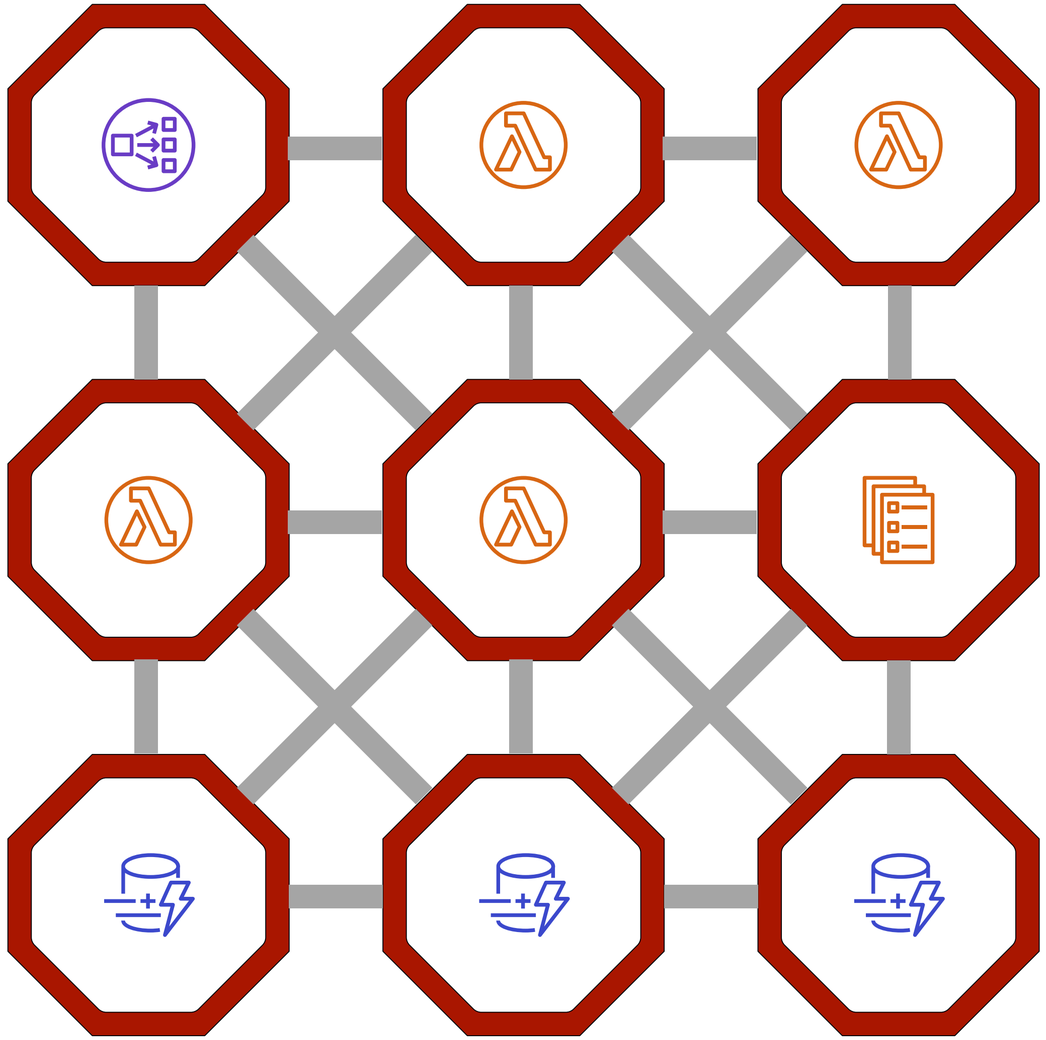
Applied Graph Theory

Managing Virtual Topologies:

CRUD Service

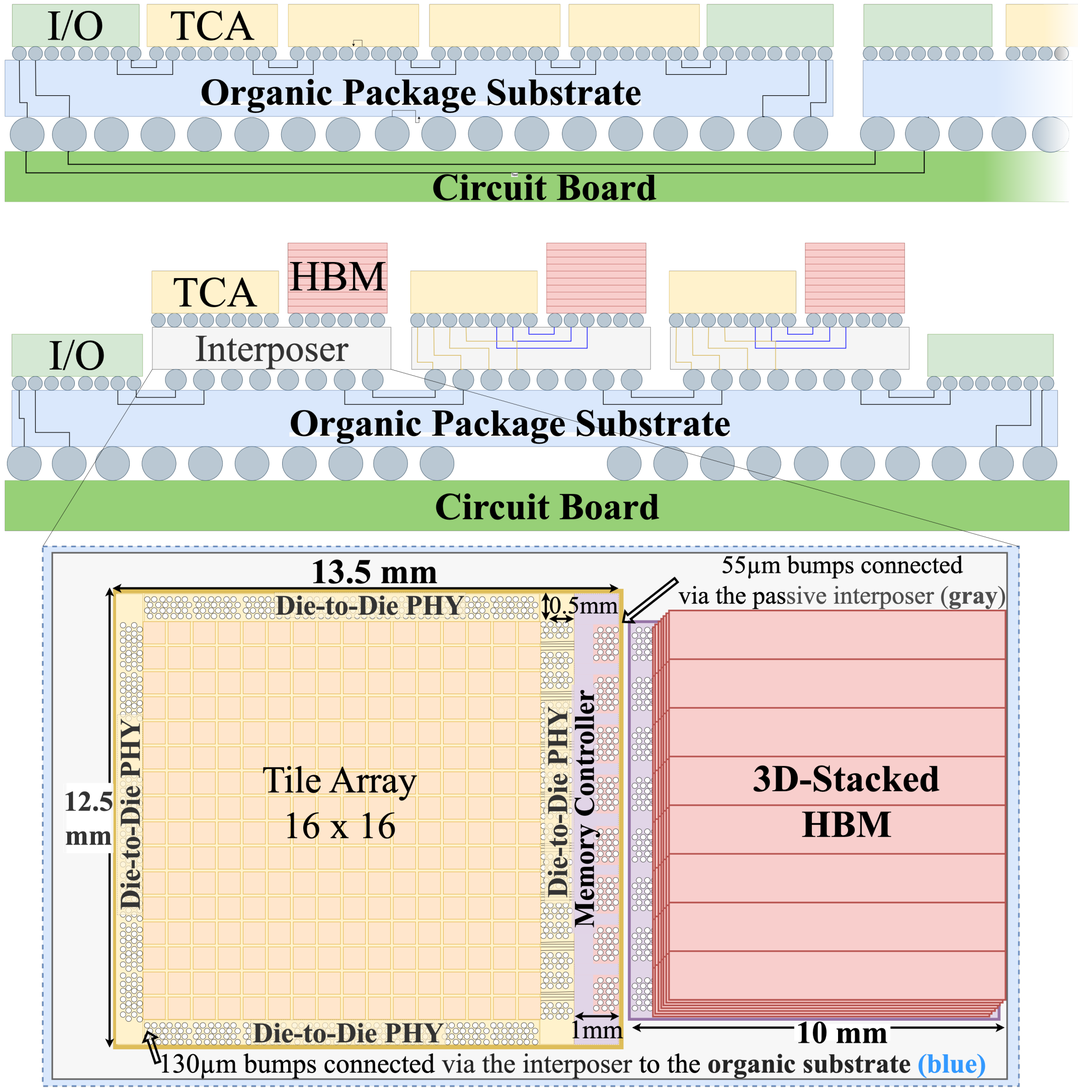
- Load Balancers
- Lambda's
- Storage
- AI/ML

- Endpoints addressed by name, rather than geographically by IP route summarization
- Graph Virtual Machine (GVM), life *beyond* Kubernetes



Massive Data Parallelism in 2.5D, 3D or 3.5D

- How do we reconcile age old design of protocols using a Newtonian Background of time, with what modern physics knows about the nature of time?
- UCIe leads the way with Load/Store but distributed systems folks need *swap*, to guarantee truly reversible (exactly once) semantics



Questions?

DÆDÆLUS addresses fundamental problems in distributed systems using Protocols, Data Structures and Algorithms Inspired by Quantum Information Theory & Multiway Systems

- We Design, Simulate and Build and Deploy Software Infrastructures for Chiplet based Microdatacenters, targeting secure, reliable, distributed computing on the edge
- Infrastructure is Topology Management: Applied Graph Theory.
  - Virtual Networks on top of Logical Networks, on top of Physical Networks
- Initial use-cases include Transaction systems, Digital Twins, AI/ML/LLM infrastructure, AR/VR, and Interface to Quantum Computers

We usually stop here to field questions.
The backup slides are available to answer the most common of these questions

Questions?

Backup Slides

How would you like to formally verify, model,
and bring-up your chiplet systems to confirm their properties?

Our tools are ready to begin, are you?

paul@daedaelus.com @plborrill
dugan@daedaelus.com @duganHammock

Chiplet Mesh Networking

- GREY FAILURE NEEDS OUR HELP

- KUBERNETES NEEDS OUR HELP

- SECURE ENCLAVES NEED OUR HELP

- THE CAP THEOREM NEEDS HELP

- PROGRAM WITH LINGUA FRANCA

- TRANSACTIONS NEED OUR HELP

Products for Distributed Systems Developers

Chiplet Fabrix:

A separate compute realm, sandwiched between the CXL bus and Ethernet, to support reliable transactions. We eliminate CAP Theorem tradeoffs by providing the illusion of an unbreakable network: detecting, isolating and healing failures far faster than protocol or application stacks using traditional timeouts and retries

Theseus: Scouting Protocols

Explore local environments to discover & retrieve knowledge of resources, constraints, and topologies in local (Chiplet) environments. Silently monitor local connectivity, device and software liveness — raising alerts when links become flakey or server software hiccups

Ariadnie: Routing Protocols

Dynamic construction and tear down of communication graphs — for consensus, load balancing and failover. Enables: observability on demand, fault isolation and distributed fault Isolation

Icarus: Secure Distributed Debugging

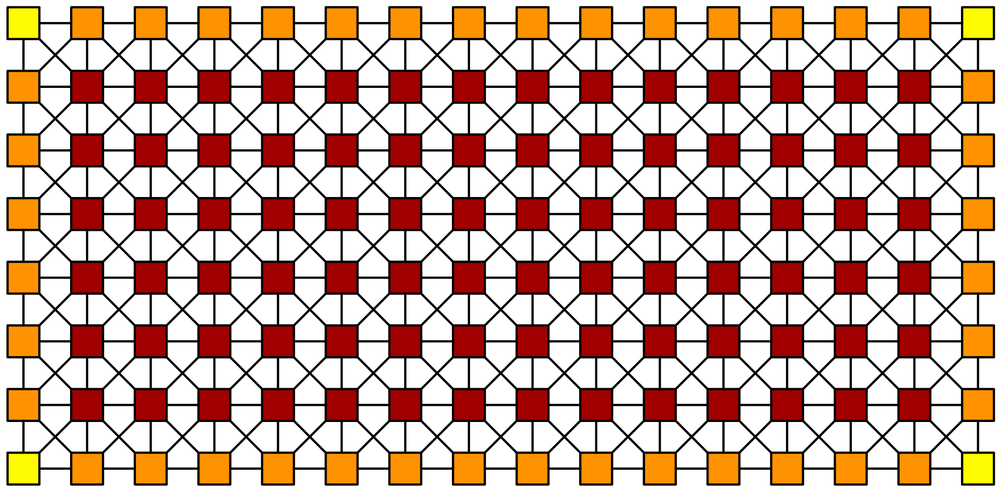
Connects secure internal world of the Chiplet Fabrix with hostile external world of legacy networks; using compositional (zero knowledge) techniques: formally verified APIs, comprehensively tested implementations

Labyrinth: Simulator

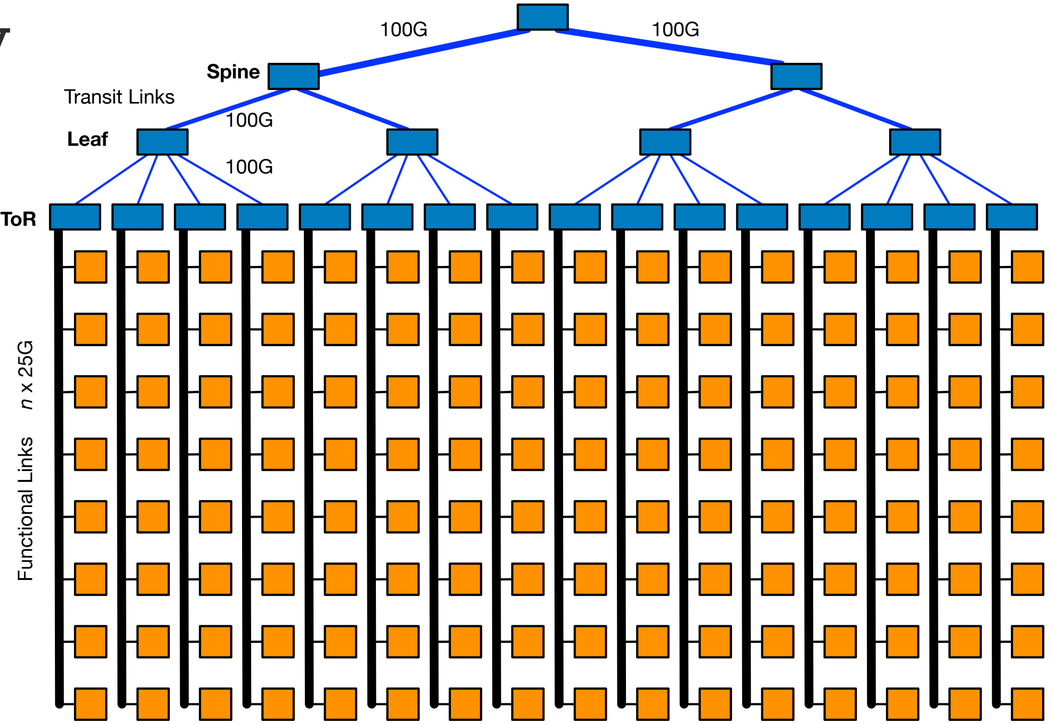
Simulator driven toolset for Chiplet based microdatacenters. Based on algorithms whose assumptions about causality go beyond simplistic notions of Newtonian time. Seeking collaboration with Antithesis and Xronos

Physical Topology: Chiplet vs. Racks

- Manage distributed systems software with Graphs, not Lists
- Comparison: Latency, Cost, Bandwidth - Chiplet Mesh wins on all fronts
- Nodes are the same (Host Processor + SmartNIC)
- Use all 8 lanes of the NIC independently
- Clos ‘combines’ multiple lanes for BW



Future Architecture



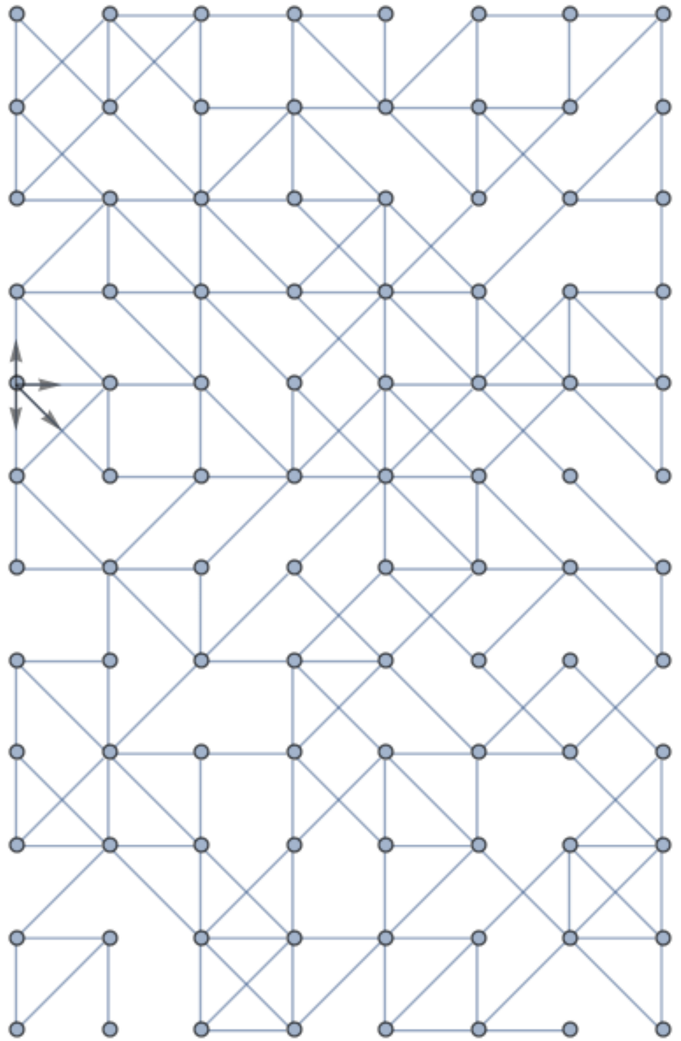
Today’s Architecture

Simulation Tools for Network Observability

Chiplet Mesh Networking:

- **Physical:** What you wake up with and discover
- **Logical:** What you provision, and dynamically respond to
- **Virtual:** What you program, monitor and interact with

- Simulation tools for network diagnostics, analysis, and planning
- Near-term focus is simulating AI/ML/LLM workloads on Chiplet Mesh Networks
- Simulation tools are working. We showed them at the Chiplet Summit



Chiplet Mesh Management

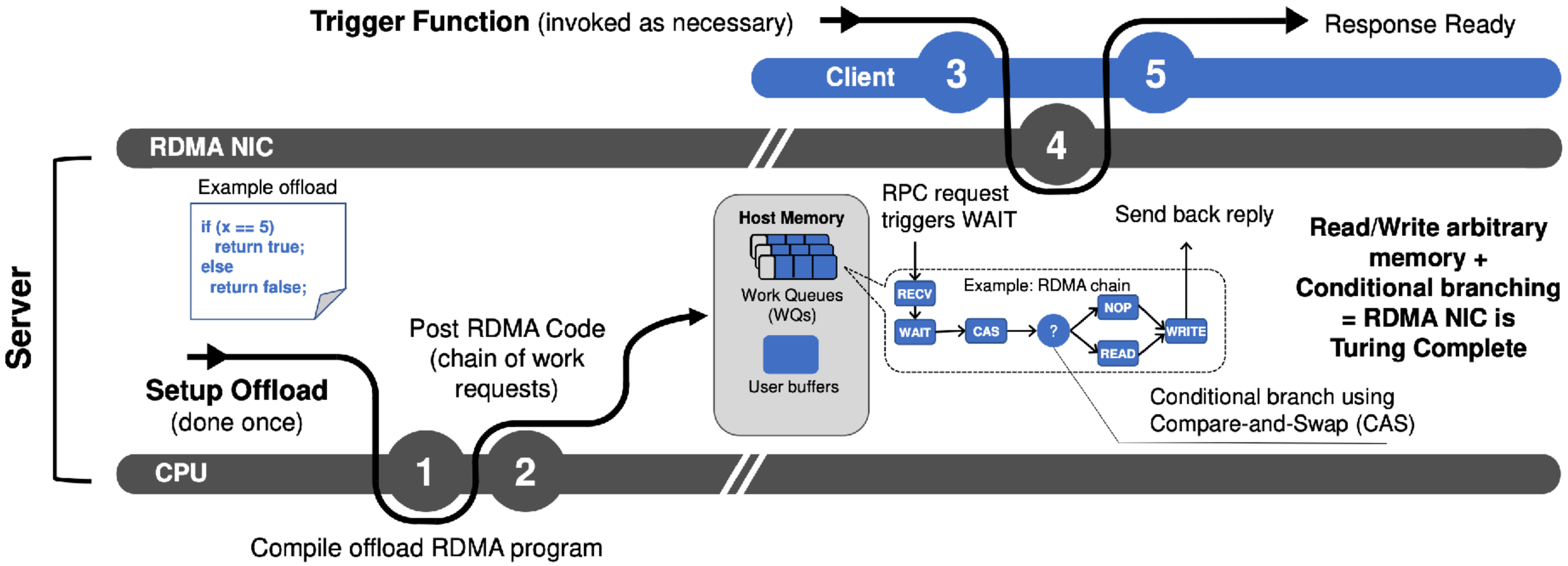
Seed Questions for Customers

- How can we cost effectively make a network connection to each of a sea of palm sized (OAM) modules? What is the right way to think about this? Backplanes, motherboards (UBB), and cables are expensive!
- How can we make management software's job easier looking at that sea of modules?
- How can we keep network addressing of endpoints within those modules doable, especially managing constant change from fail/restart, migrations, etc?
- How can we help distributed applications keep application state consistent when it's divided among an order of magnitude more servers?

- We need questions from Customers. They're more interesting than ours!

RDMA is FITO*

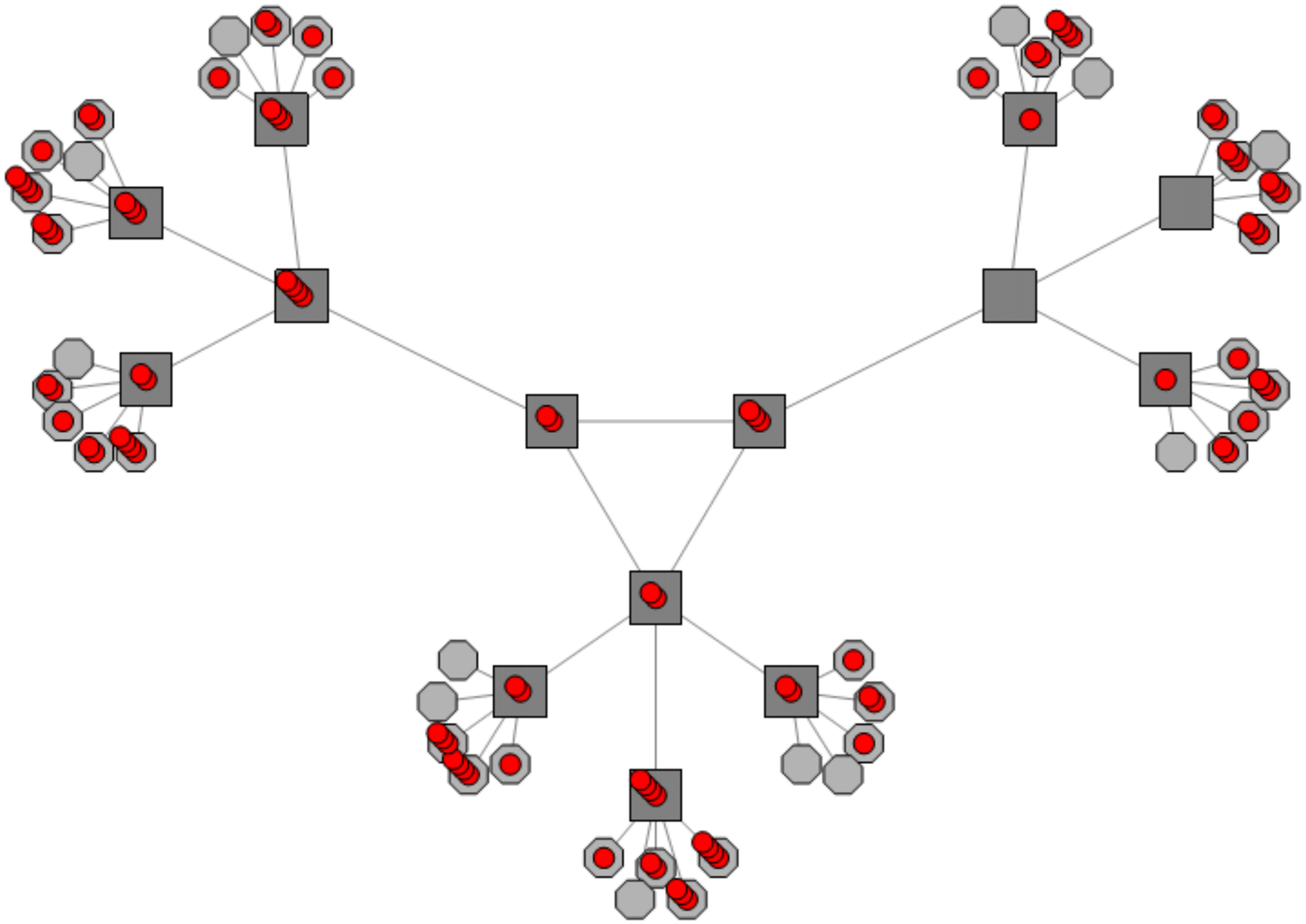
*Forward-In Time Only thinking



From: [RDMA Is Turing Compete. We Just Didn't know it yet]

Network Simulation (Clos topology) (100 Packets)

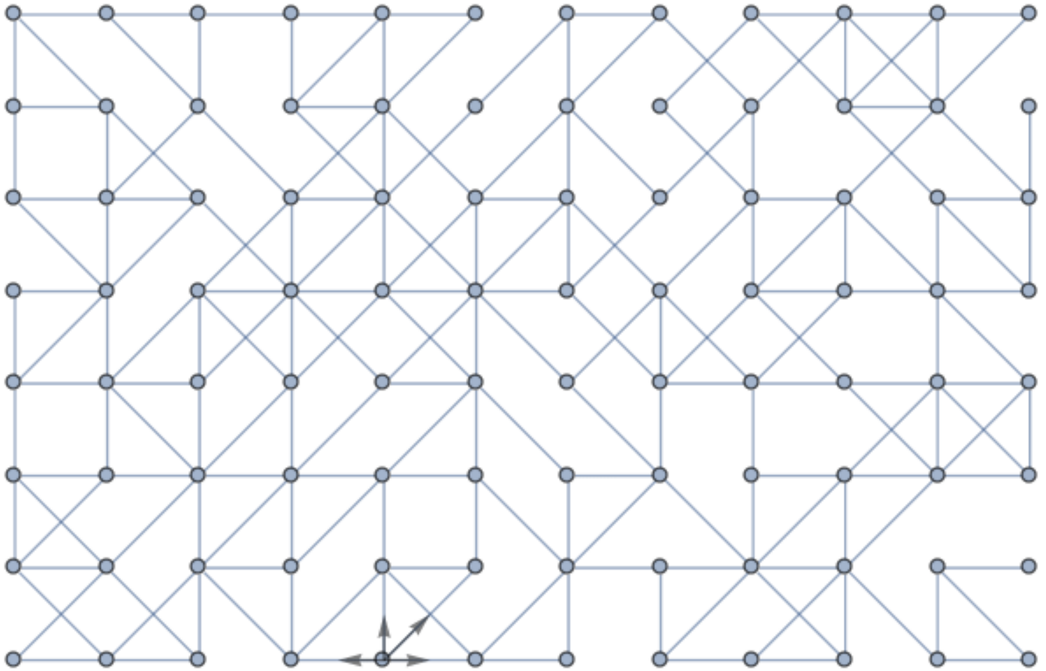
- This is what a Clos looks like from a different graph perspective.
- All servers are leaves. They don’t participate in forwarding
- All the switches are roots for the spanning trees, they force the applications to be oblivious to network perturbations



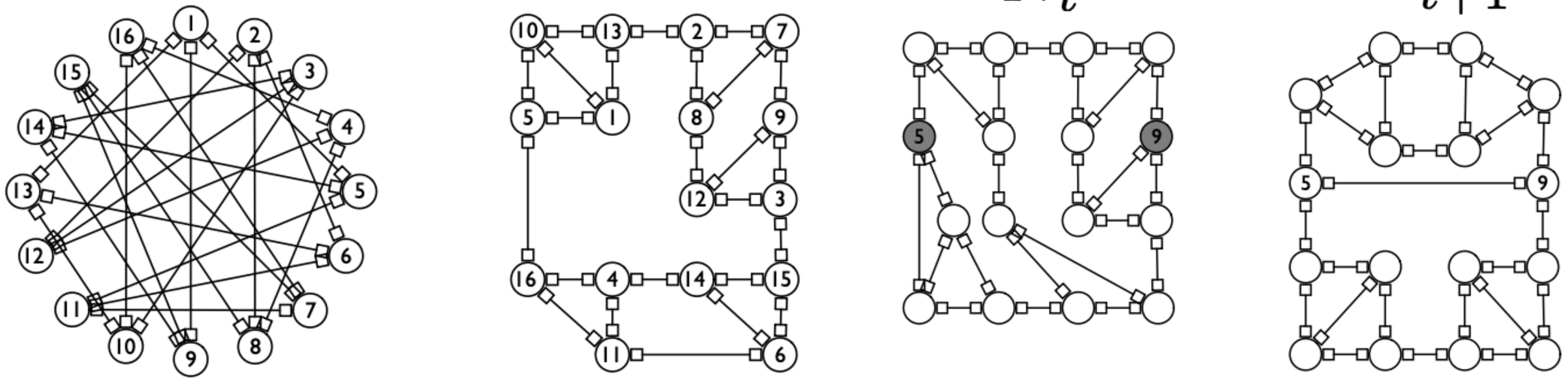
Chiplet Fabrix Simulation Tools

Chiplet Mesh Networking

- **Explore:**
- **Physical:** What you wake up with and discover
- **Logical:** What you provision, and dynamically respond to
- **Virtual:** What you program, monitor and interact with



What Else can you do with Expander Networks?



(a) Network: demand-oblivious (b) Network: demand-aware (c) Network: self-adjusting

Virtual Networks

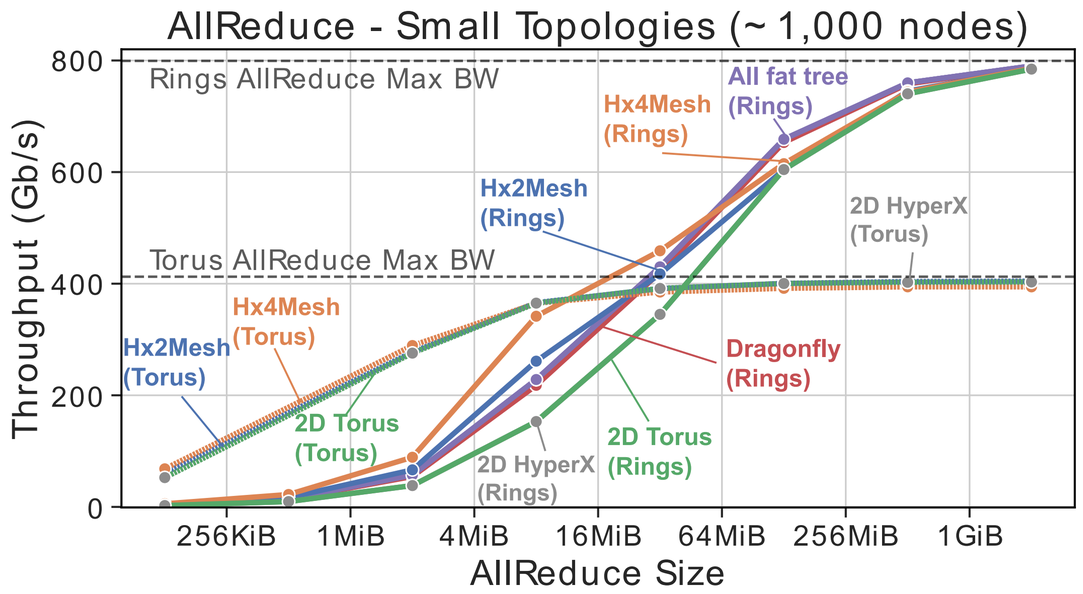
On top of **Logical Networks**

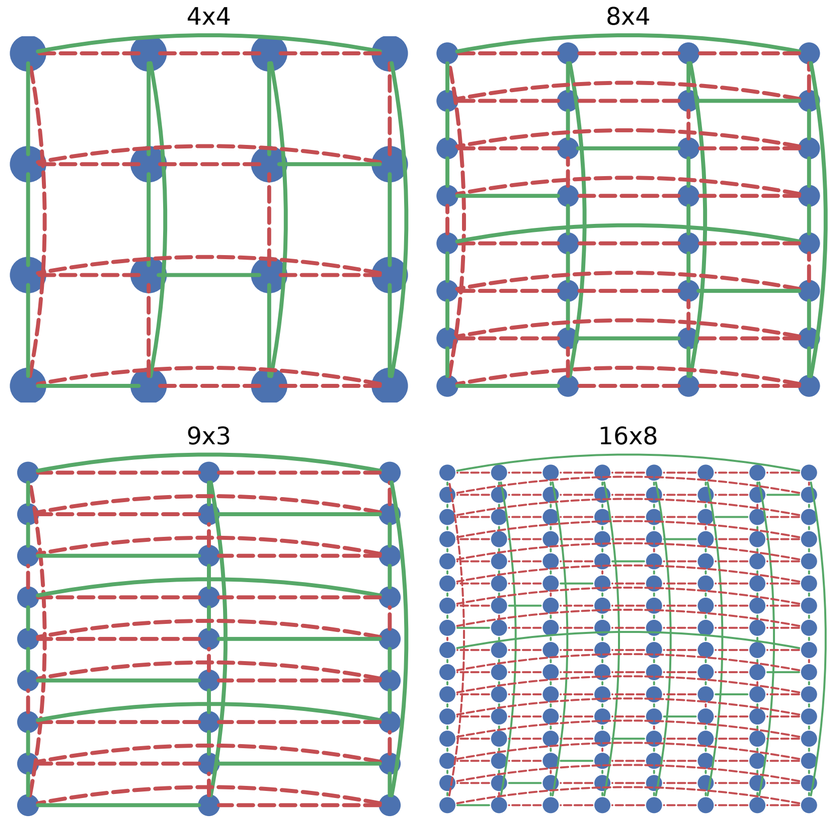
On top of **Physical Networks. (Ethernet Mesh)**

Graph Programmable Connectivity

for Rack Scale & Chiplet Based Microdatacenters

- Can we make Chiplet Infrastructures auto-configuring?
- Can we make Chiplet Infrastructures self-healing?
- Can we design for a 100 year lifetime?





HammingMesh

Cost and Bandwidth Comparison

Topology	Small Cluster (≈1,000 accelerators) cost [M$]	glob. BW [% inject]	global saving	ared. BW [% peak]	ared. saving	diam.	Large Cluster (≈16,000 accelerators) cost [M$]	glob. BW [% inject]	global saving	ared. BW [% peak]	ared. saving	diam.
nonbl. FT	25.3	99.9	1.0x	98.9	1.0x	4	680	98.9	1.0x	99.8	1.0x	6
50% tap. FT	17.6	51.2	0.7x	98.9	1.4x	4	419	47.6	0.8x	99.8	1.6x	6
75% tap. FT	13.2	25.7	0.5x	98.9	1.9x	4	271	24.0	0.6x	99.8	2.5x	6
Dragonfly	27.9	62.9	0.6x	98.8	0.9x	3	429	71.5	1.2x	98.6	1.6x	5
2D HyperX[2]	10.8	91.6	**2.1x**	98.1	2.3x	4	448	95.8	1.5x	91.4	1.4x	8
Hx2Mesh	5.4	25.4	1.2x	98.3	4.7x	4	224	25.0	0.8x	92.3	2.8x	8
Hx4Mesh	2.7	11.3	1.0x	98.4	**9.3x**	8	43.3	10.5	**1.7x**	92.2	**14.5x**	8
2D torus	2.5	2.0	0.2x	98.1	10.1x	32	39.5	1.1	0.2x	91.4	15.7x	128

TABLE II: Overview of our example networks (small and large cluster) using the cost model in Section III-C. All bandwidths are the result of the packet-level simulations detailed in Section V-A. Global alltoall bandwidth is reported as share of the injection bandwidth for large messages (1.6 Tb/s). Allreduce bandwidth is reported as share of the theoretical optimum (1/2 of the injection bandwidth) for large messages. The cost savings for global and allreduce bandwidth are relative to the corresponding network cost of the nonblocking fat tree. Note that the diameter counts all cables and its derivation is explained in Section III-B.

- **Mobile Execution environments** (Containers & Lambdas) make network diameters *arbitrarily small* (one hop is most common for AI/ML relationships)

Objections

The Strawman objection, by conventional network architects, uses ‘diameter’ as a figure of merit (or demerit). This exposes three misconceptions:

(1) Nodes are “fixed” in some Cartesian layout, nailed to the floor of the datacenter in some specific rack **(our addressable entities are mobile)**

(2) Hops are somehow related to latency in a fixed cartesian coordinate system, so the diameter gets worse as you scale **(we exploit neighbor locality)**

(3) Execution entities are randomly placed throughout the datacenter — for a ‘flat (oblivious) network’ concept. **(Ours are naturally clustered on deployment, AND can be migrated toward each other with a spring-electrical embedding in the orchestration protocols)**

Mesh Networks Map AI/LLM computations

- Interconnect
  - Physical is 8-ports.
  - Logical 6 ports
  - Virtual 4 ports
- Maps directly to AI/LLM Topology

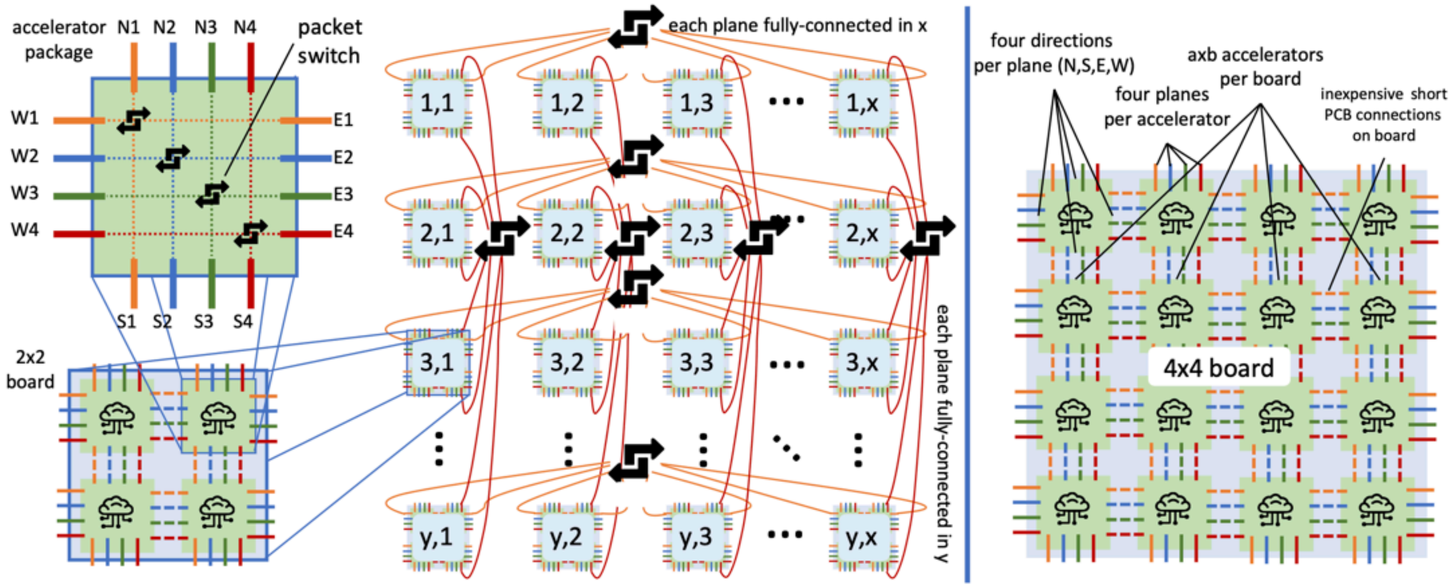


Fig. 3: HammingMesh structure: left $x \times y$ Hx2Mesh, right Hx4Mesh board, both with four planes.

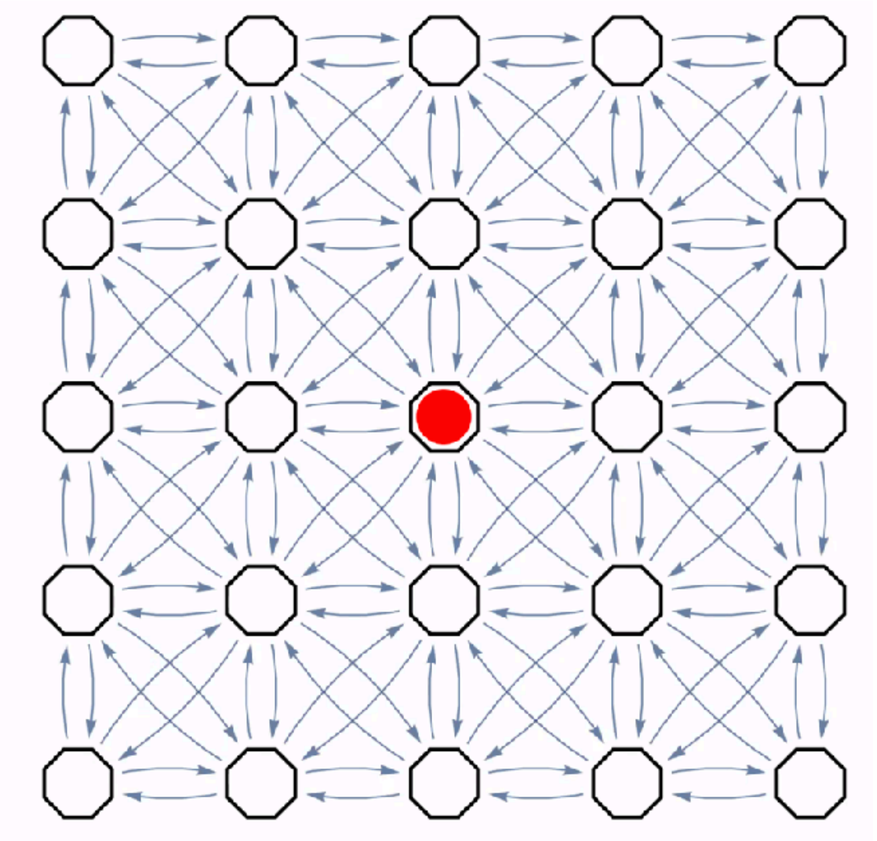
Topology	Small Cluster (≈1,000 accelerators) cost [M$]	glob. BW [% inject]	global saving	ared. BW [% peak]	ared. saving	diam.	Large Cluster (≈16,000 accelerators) cost [M$]	glob. BW [% inject]	global saving	ared. BW [% peak]	ared. saving	diam.
nonbl. FT	25.3	99.9	1.0x	98.9	1.0x	4	680	98.9	1.0x	99.8	1.0x	6
50% tap. FT	17.6	51.2	0.7x	98.9	1.4x	4	419	47.6	0.8x	99.8	1.6x	6
75% tap. FT	13.2	25.7	0.5x	98.9	1.9x	4	271	24.0	0.6x	99.8	2.5x	6
Dragonfly	27.9	62.9	0.6x	98.8	0.9x	3	429	71.5	1.2x	98.6	1.6x	5
2D HyperX[2]	10.8	91.6	**2.1x**	98.1	2.3x	4	448	95.8	1.5x	91.4	1.4x	8
Hx2Mesh	5.4	25.4	1.2x	98.3	4.7x	4	224	25.0	0.8x	92.3	2.8x	8
Hx4Mesh	2.7	11.3	1.0x	98.4	**9.3x**	8	43.3	10.5	**1.7x**	92.2	**14.5x**	8
2D torus	2.5	2.0	0.2x	98.1	10.1x	32	39.5	1.1	0.2x	91.4	15.7x	128

This is misleading

Scouting and Routing on An Ethernet Mesh

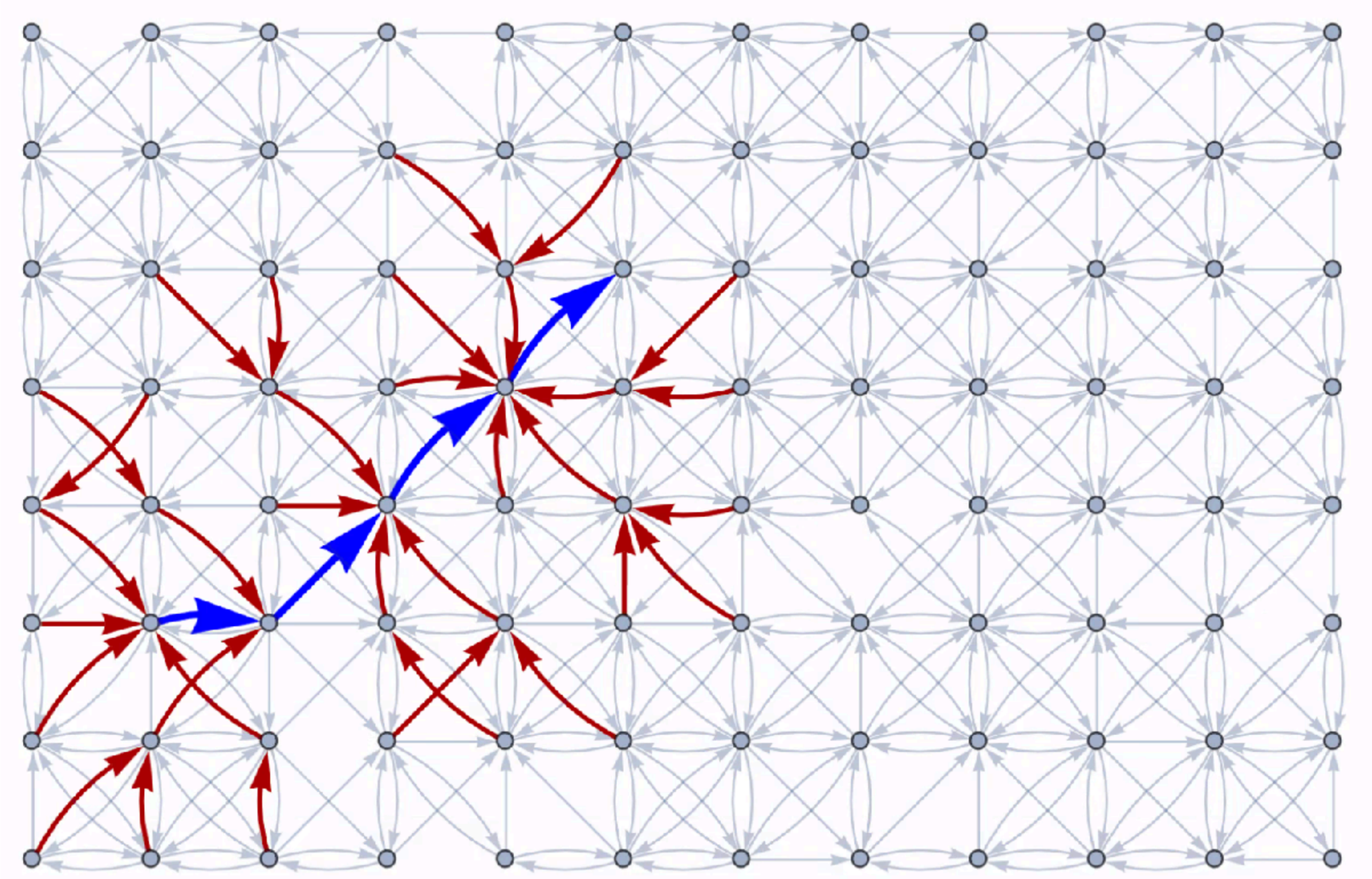
Boot-up assumptions:

- Octagonal Chiplet Substrate
- Packets explore the local environment and “discover” their topology
- All 8 lanes are used independently
- Graph Theory connects Hexagonal graphs to 8-port graphs in a “3rd” dimension
- Octrees are used in computer graphics and extend the concept of quadtrees into 3D. Hex trees also exist but are less common



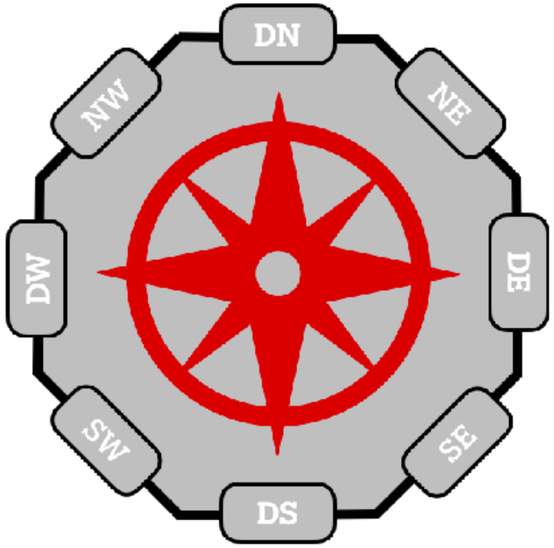
Scouting and Routing on An Ethernet Mesh

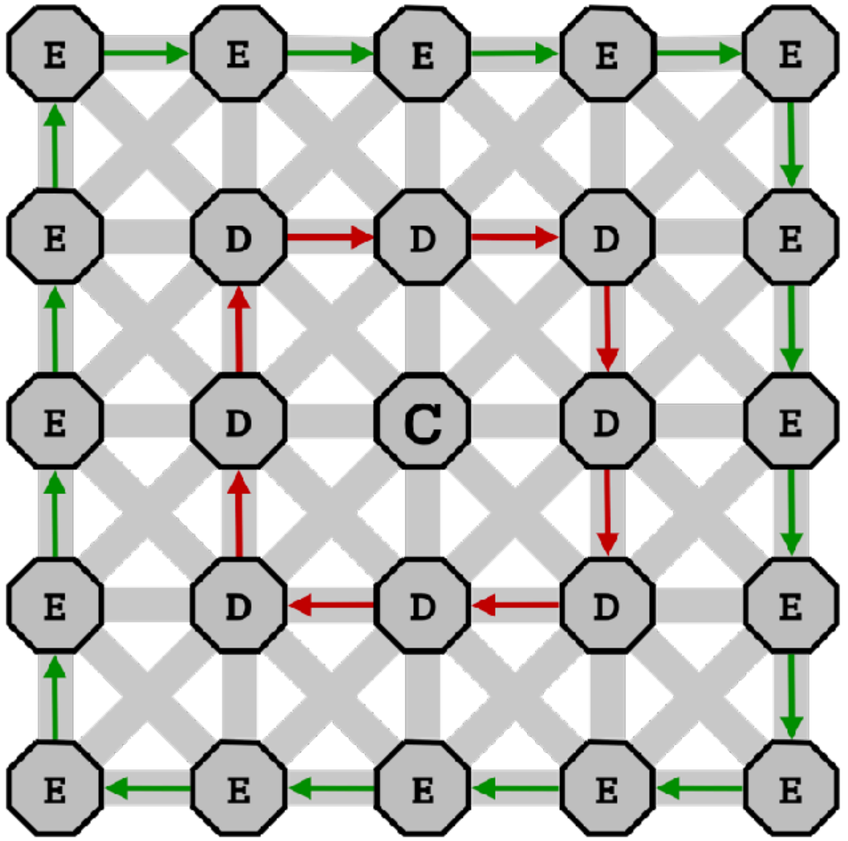
- Theseus: Scouting the Labyrinth



Topology Logical Neighborhoods

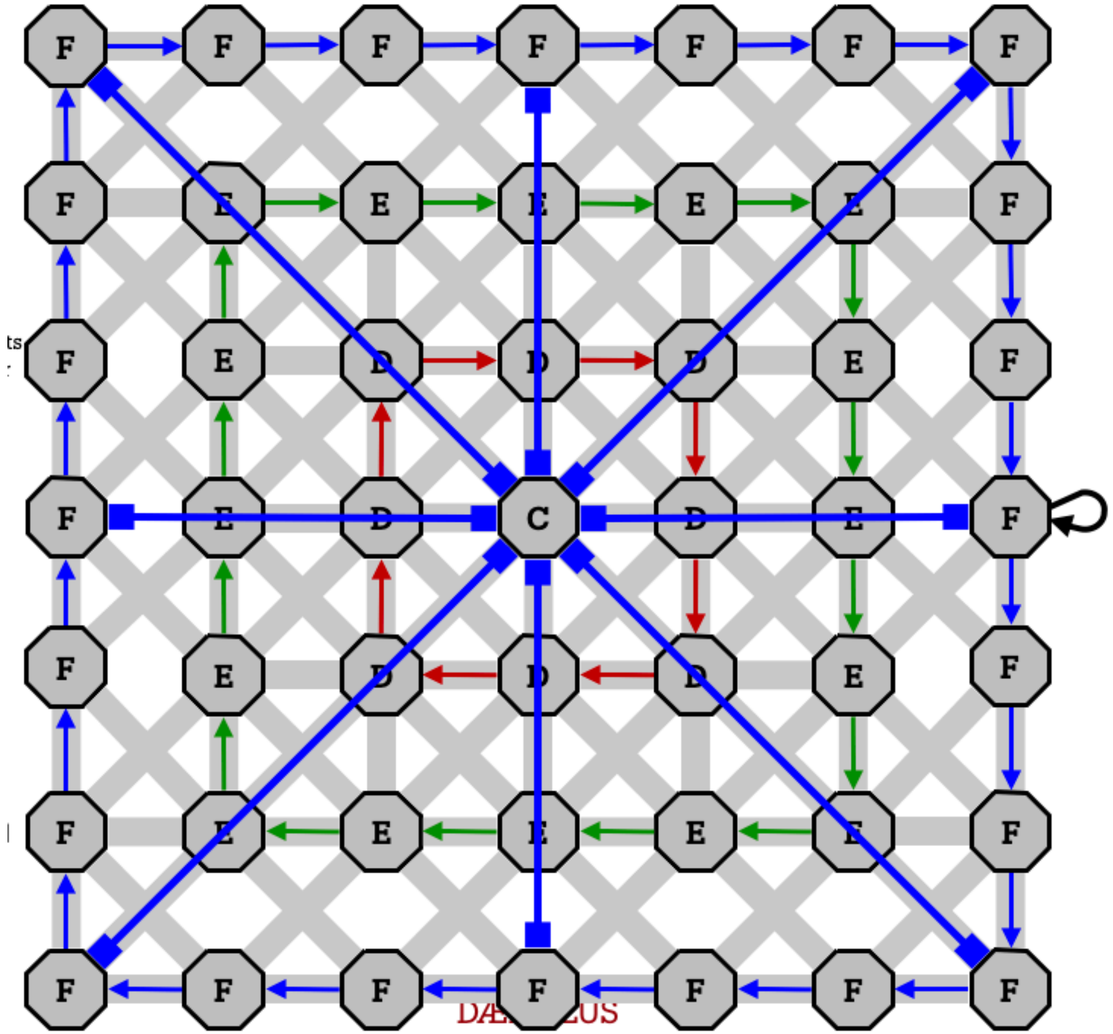
- **Discovery**
  - Checking out the Labyrinth
  - Compass Point Scouting for local topology
  - Routing Trees for global connectivity





Exploring the Boundaries

- **Ariadne**
  - Checking out boundaries with a Magic Thread
  - Setting up Trees



Key Technical Messages

1. We can't recover lost information at Layer 3 (IP) or layer 4 (TCP) if events were thrown away (are not paired) at L2.
2. We can't solve causal determinacy unless we have a conserved quantities mechanism in the bits on the wire protocol.
3. We can use applied graph theory independently in the Physical, Logical and Virtual layers to manage computational resources.

In the near future, We will incorporate indefinite causality mechanisms in the FPGA substructure

- Alternating Causality (AC) protocols, in SmartNIC FPGAs provide mechanisms to manage non-monotonic behaviors (hiding side channels), while presenting to an observer on the other side of the CXL bus, an illusion of logical monotonicity that does not corrupt data structures.
- SmartNIC FPGAs operations identified this issue (reads and writes) explicitly. The UCIe protocols are load/store only.
- We discovered we can't achieve exactly once semantics without reversible subtransactions, and we can't have reversible subtransactions without swap (and unswap — causal reversal)
- This is part of the proposal we would like to present to the UCIe and CXL experts

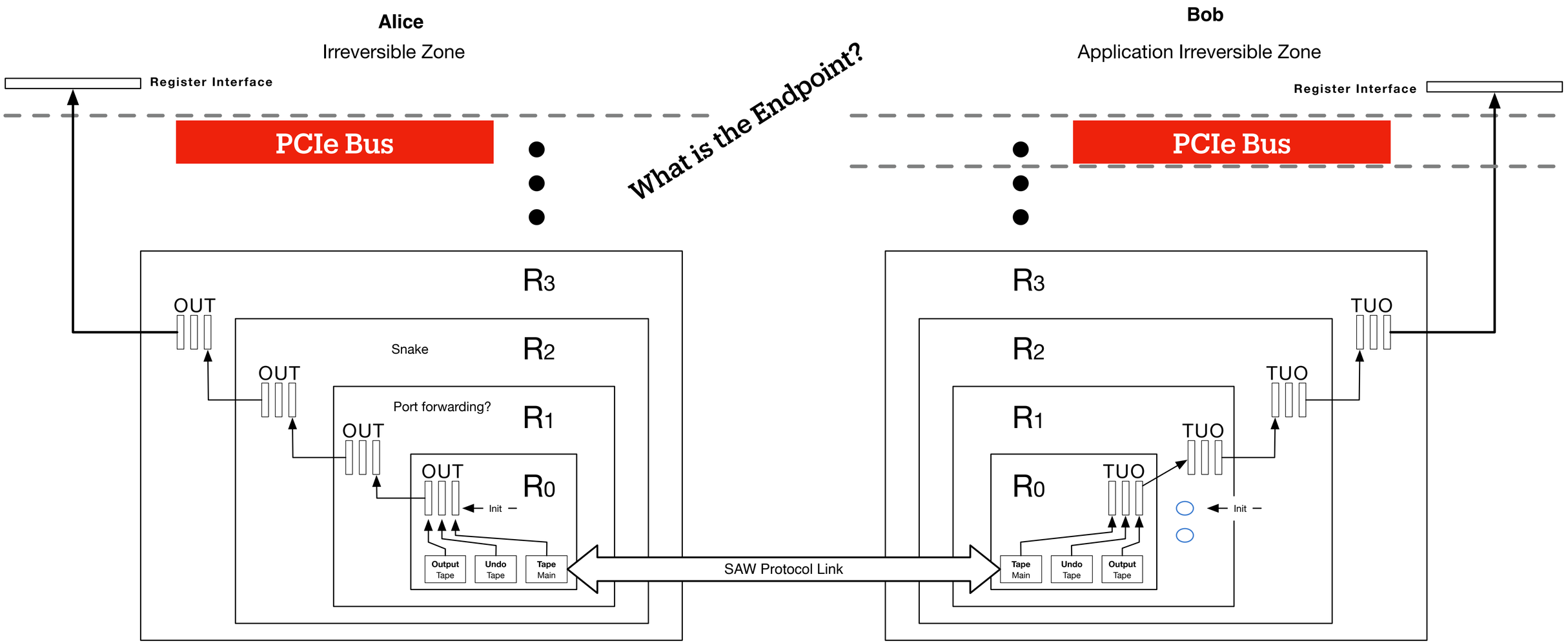
Logging and locking with Ariadne

LOGGING AND LOCKING: Another solution

Logging and locking are an alternative implementation of the transaction concept. The legendary Greeks, Ariadne and Theseus, invented logging. Ariadne gave Theseus a magic ball of string which he unraveled as he searched the Labyrinth for the Minotaur. Having slain the Minotaur, Theseus followed the string back to the entrance rather then remaining lost in the Labyrinth. This string was his log allowing him to undo the process of entering the Labyrinth. But the Minotaur was not a protected object so its death was not undone by Theseus' exit.

Hansel and Gretel copied Theseus' trick as they wandered into the woods in search of berries. They left behind a trail of crumbs that would allow them to retrace their steps by following the trail backwards, and would allow their parents to find them by following the trail forwards. This was the first undo and redo log. Unfortunately, a bird ate the crumbs and caused the first log failure.

Successive Reversibility



Dual Petri Spekkens Protocol

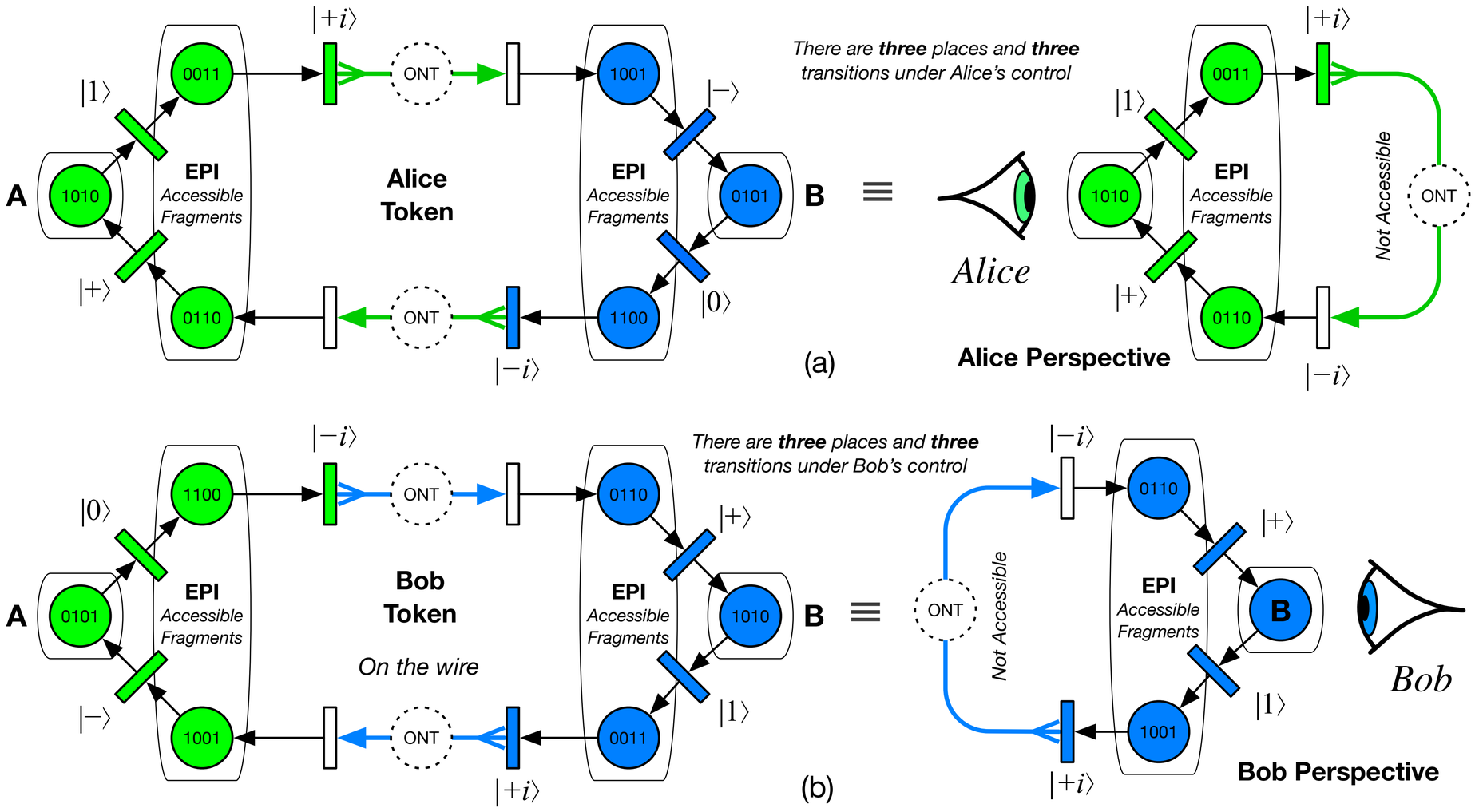
Interlocked Send & Receive
Guaranteed in the sense that both sides "know" if it failed or succeeded

- *New* protocol
- Ethernet Frames

- Reversible
- No-Copying

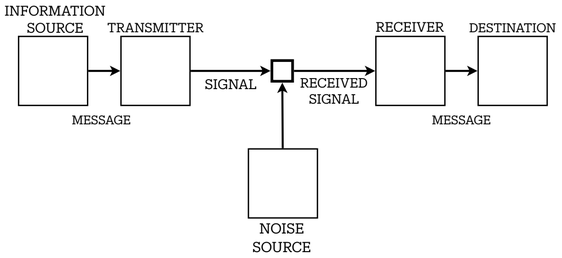
- First class object
- Compositional

Based on Quantum Ethernet



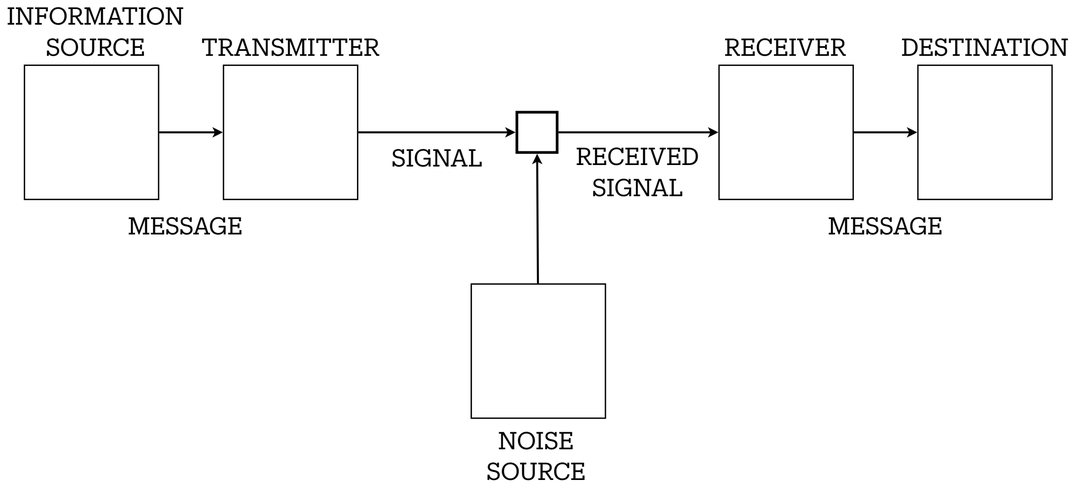
Guaranteed in the sense that both sides 'know' if it failed or succeeded

Fundamentals: Shannon Channel



This is an Animated Story

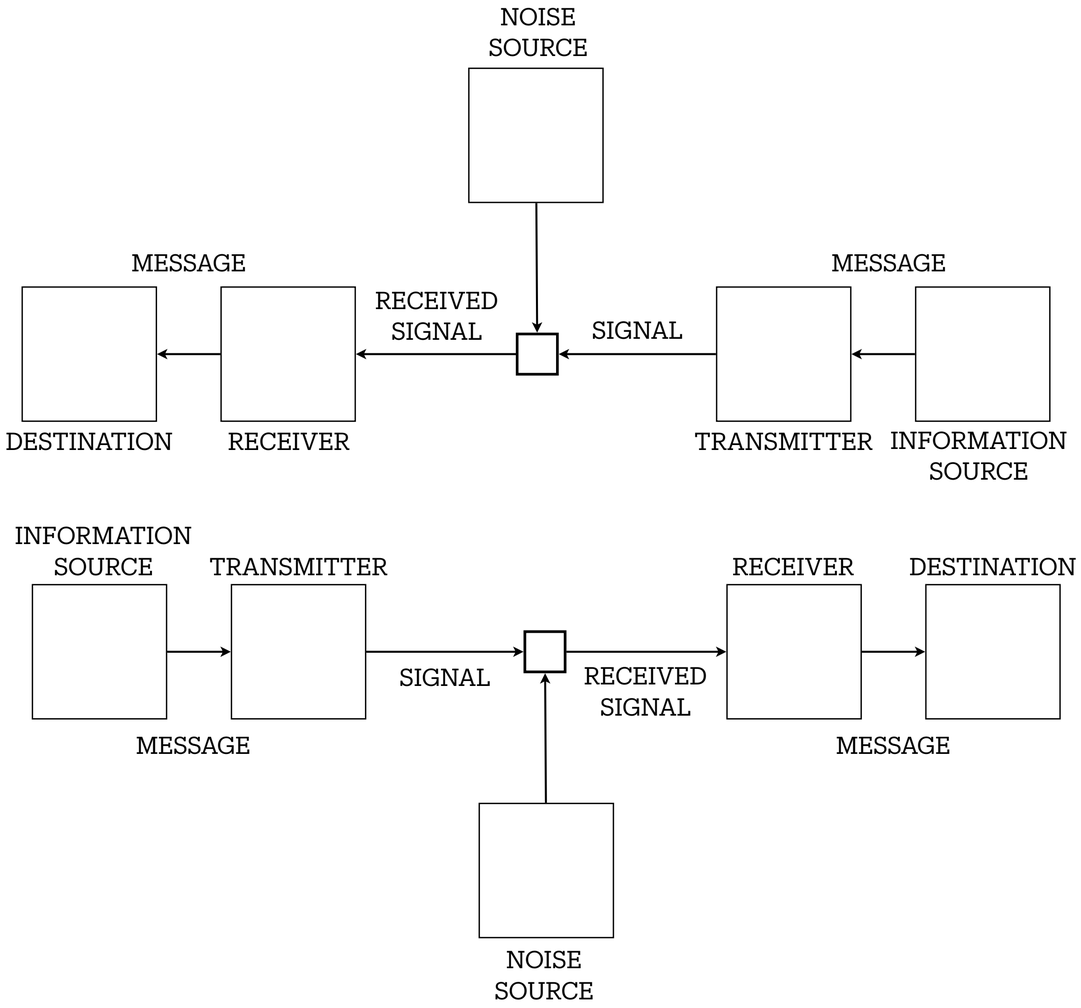
General Communication System



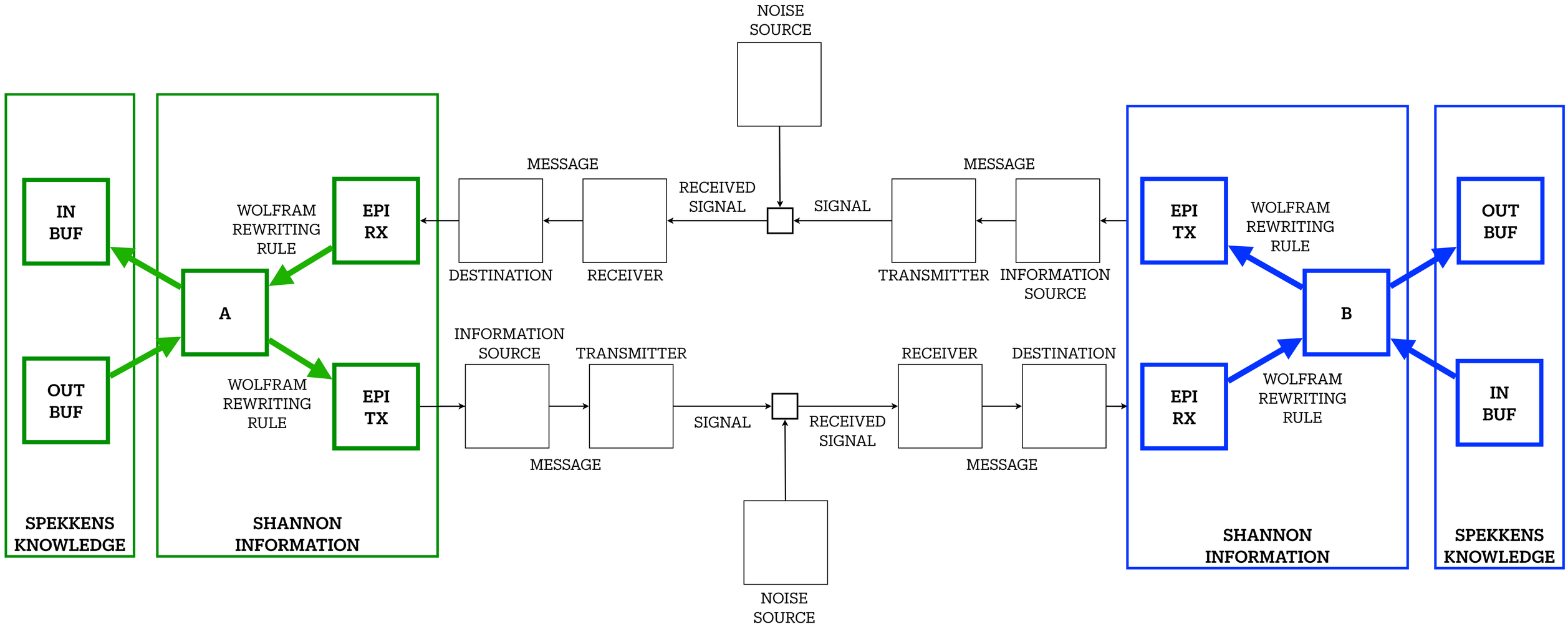
A Mathematical Theory of Communication

By C. E. SHANNON.

General Communication System



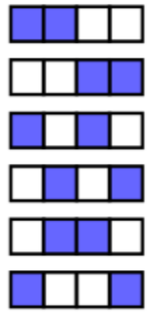
General Communication System



STEALING (SWAP) STATE

The Combinatorics of Entanglement

Knowledge is captured information - Events in the SerDes



Connecting: Spekkens & Hardy (Quantum) with Moses & Halpern (Computer Science)

Knowledge Balance Principle (KBP) Spekkens [1]

- If one has maximal knowledge, then for every system, at every time, the amount of knowledge one possesses about the ontic state of the system at that time must equal the amount of knowledge one lacks

Generalized Reversibility Model

- Determinism comes from Reversibility. We Are Missing Half The Picture. Hardy [2,3]
- Counterfactuals. Marletto [4]

Common Knowledge in Computer Science

- Knowledge and Common Knowledge [5], Common Knowledge Regained [6]

	$\{\|0\rangle, \|1\rangle\}$	$\{\|+\rangle, \|-\rangle\}$	$\{\|+i\rangle, \|-i\rangle\}$
$\|\Phi^+\rangle$	C	C	A
$\|\Phi^-\rangle$	C	A	C
$\|\Psi^+\rangle$	A	C	C
$\|\Psi^-\rangle$	A	A	A

$$1 \vee 2 \Leftrightarrow |0\rangle$$
$$3 \vee 4 \Leftrightarrow |1\rangle$$
$$1 \vee 3 \Leftrightarrow |+\rangle$$
$$2 \vee 4 \Leftrightarrow |-\rangle$$
$$2 \vee 3 \Leftrightarrow |+i\rangle$$
$$1 \vee 4 \Leftrightarrow |-i\rangle$$

References:

(1) In defense of the epistemic view of quantum states: a toy theory. Robert W. Spekkens. https://arxiv.org/pdf/quant-ph/0401052.pdf
(2) Are quantum states real? Lucien Hardy. https://arxiv.org/abs/1205.1439
(3) Time Symmetry in Operational Theories. Hardy https://arxiv.org/abs/2104.00071v1
(4) The Science of Can and Can't. Chiara Marletto. https://www.chiaramarletto.com/books/the-science-of-can-and-cant/
(5) Knowledge and Common Knowledge in a Disturbuted Environment. Moses, Halpern. https://web.eecs.umich.edu/~manosk/assets/papers/p549-halpern.pdf
(6) Common Knowledge Regained. Gonczarowski, Moses. https://arxiv.org/abs/2311.04374
(7) The Combinatorics of Causality. Stefano Gogosio, Nicola Pinzani https://arxiv.org/abs/2206.08911

End

Making Transactions Reliable